

An Introduction to Computational Neuroscience

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University of Science and Technology of China

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Course website

<https://github.com/Wenlab/Computation-Neuro-Course>

📖 README

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Computational-Neuroscience-Course

This repository serves as an ongoing effort to create a systems and computational neuroscience course. I hope to develop a two-semester course for USTC biophysics students. The first semester will cover some basic materials, and the second semester will introduce more advanced topics for undergraduate and graduate students.

When: Thursday 9:45 am - 11:20 am, Fall 2024
Where: 东区第二教学楼2402
Teacher: 温泉 qwen@ustc.edu.cn
Teaching Fellow: 冼奇琪 xianqiqi@mail.ustc.edu.cn 程玉锟 cyk_phy@mail.ustc.edu.cn 李懿轩 yixuanli@mail.ustc.edu.cn

Recommended Textbooks:

- [Theoretical Neuroscience: Computational and Mathematical Modeling the Neural System](#)
- [Principles of Neural Design](#)

For a general reader:

- [Models of the Mind](#)

Course Performance Evaluation:

- Homework: 70%
- Final: 30% (take-home exam)

Prerequisite:

- High school knowledge of biology and neuroscience
- Proficient in Python/MATLAB/Julia
- Working knowledge of Multivariate Calculus, Probability Theory, Linear Algebra, and Differential Equations

Summary

🇨🇳

Languages

●

 MATLAB 71.8%

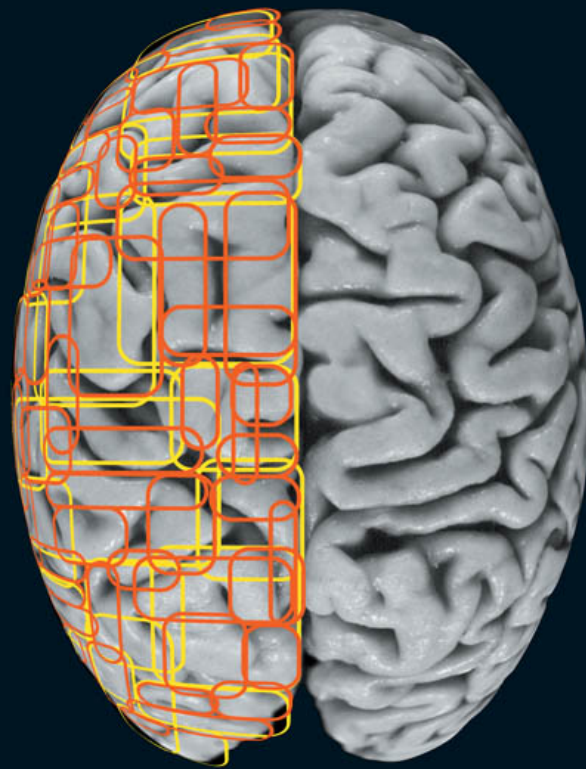
●

 Julia 28.2%

Recommended textbooks

THEORETICAL NEUROSCIENCE

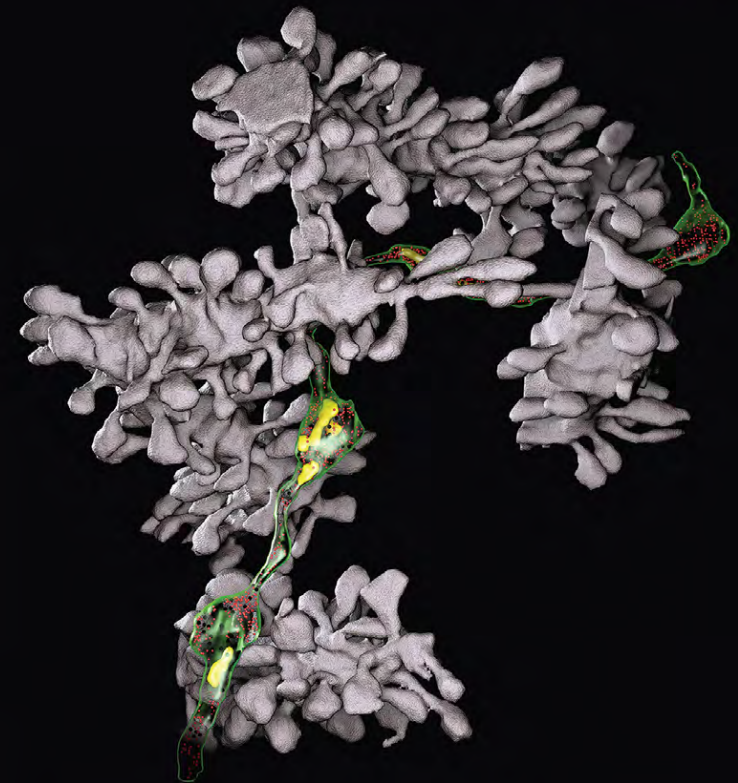
Computational and Mathematical
Modeling of Neural Systems



Peter Dayan and L. F. Abbott

How questions

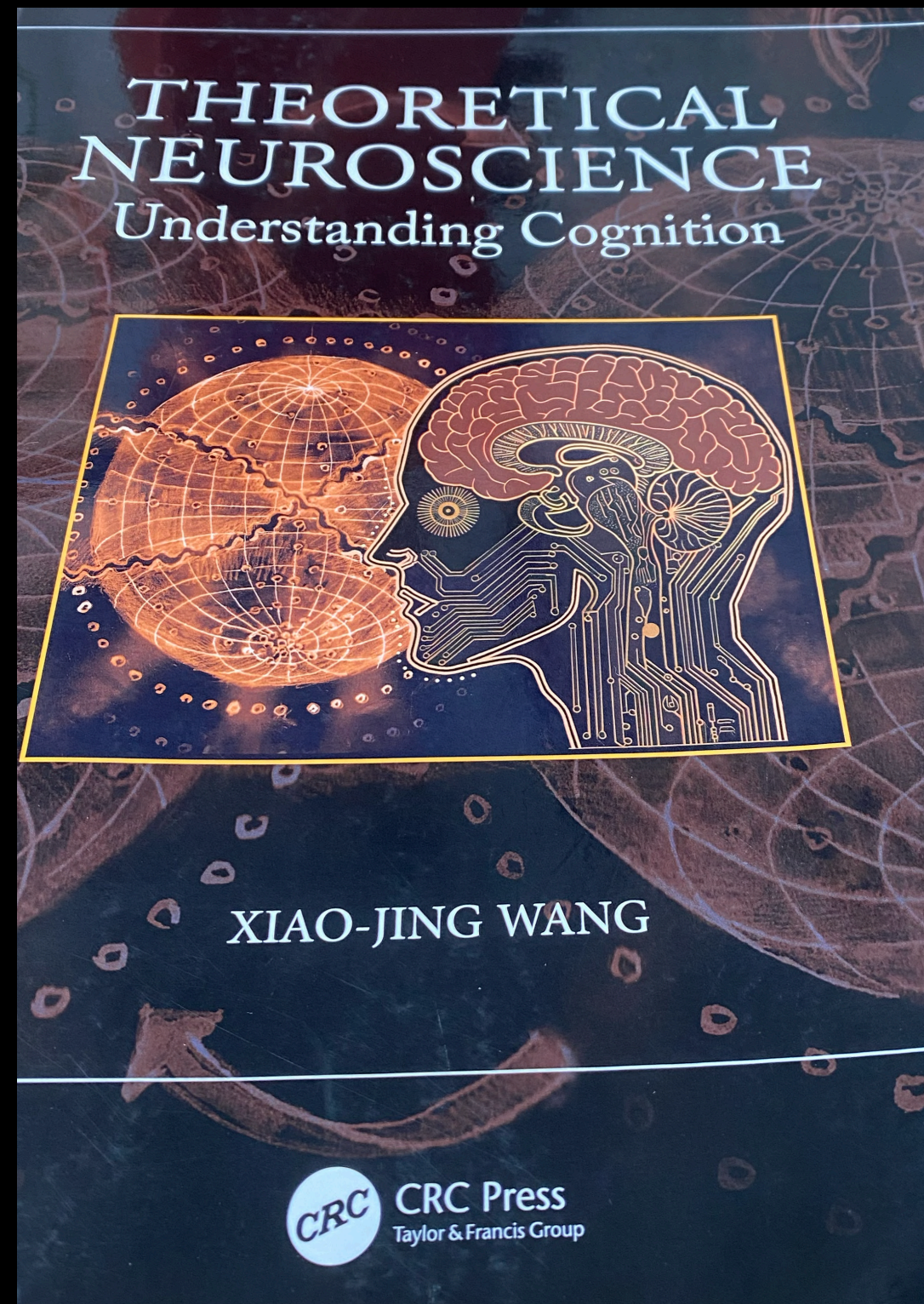
Principles of Neural Design



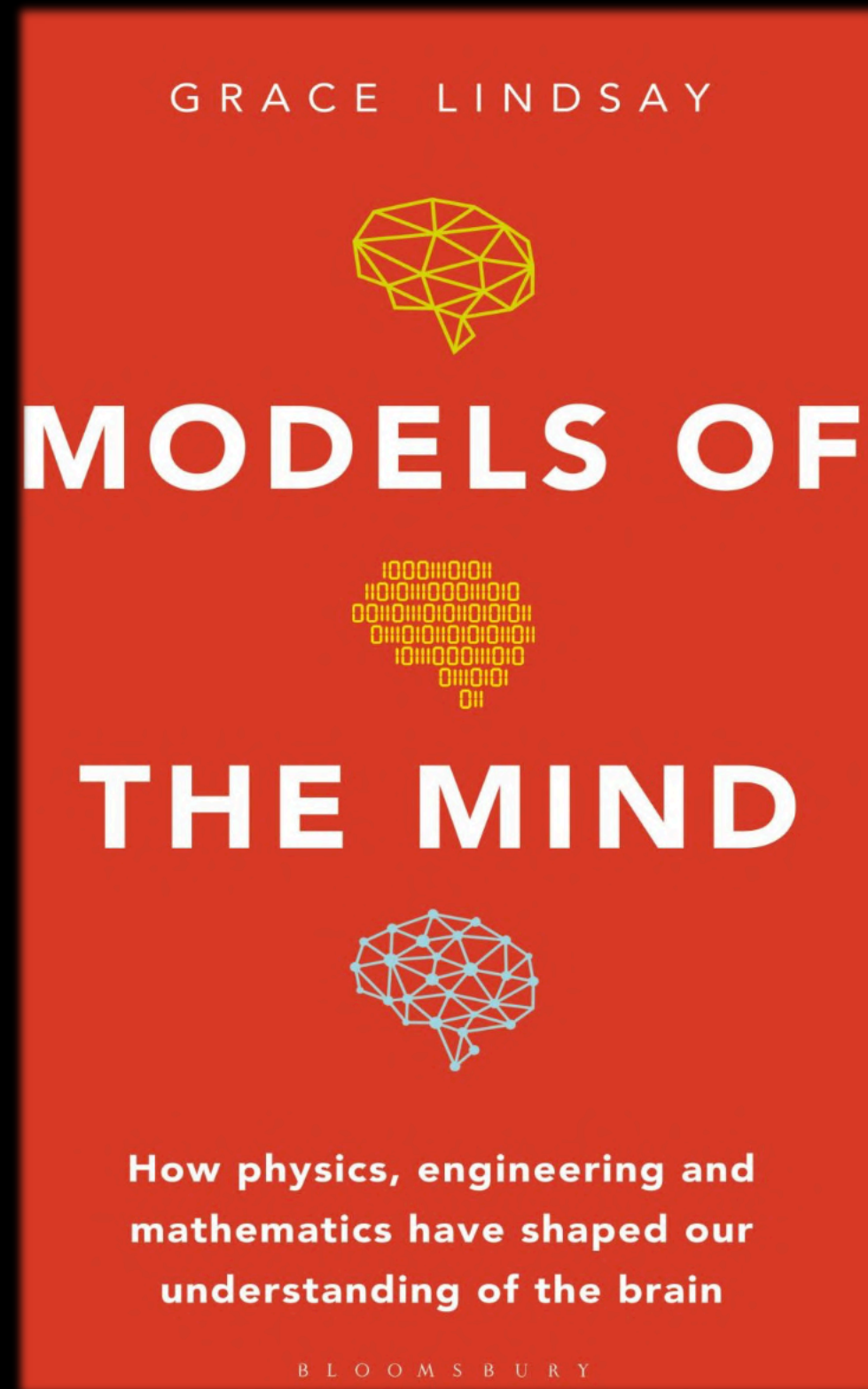
Peter Sterling and Simon Laughlin

Why questions

Recommended textbooks



Recommended book for a general audience



PRINCIPLES OF NEUROBIOLOGY

SECOND EDITION

— LIQUN LUO —



CRC Press
Taylor & Francis Group

A GARLAND SCIENCE BOOK

666 pages

SEARCHING FOR PRINCIPLES OF NEUROBIOLOGY

SECOND EDITION

— LIQUN LUO —



CRC Press
Taylor & Francis Group

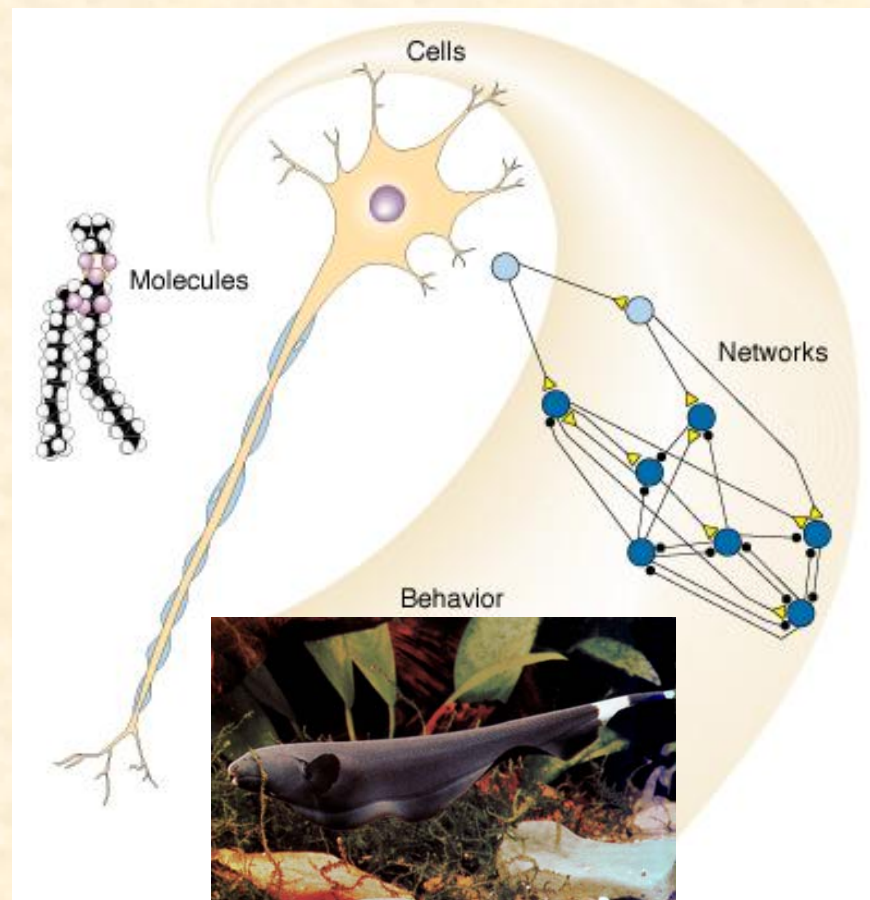
A GARLAND SCIENCE BOOK



A general impression of the brain

- The brain is squishy and messy. It may be impossible to understand the brain from a small number of principles just like in physics.
- There are too many (interesting) facts! And almost every rule has an exception.

Multiscale Organization of the Nervous System



Delcomyn 1998

Organism

1 m

System

10 cm

Map

1 cm

Column

1 mm

Neuron

10 μm

Synapse

1 μm

Molecule

10 $\text{\AA}$

“To pile speculation on speculation, I would say that the next stage could be hierarchy or specialization of function, or both.... with increasing complication at each stage, we go on up the hierarchy of sciences. We expect to encounter fascinating and, I believe, very fundamental questions at each stage in fitting together less complicated pieces into the more complicated system and understanding the basically new types of behavior which can result.”

P. W. Anderson, Science 1972
condensed matter physicist

More is different

Fitzgerald: The rich are **different** from us.

Hemingway: Yes, they have **more** money.

a conversation somewhere in Paris in 1920s

“I never had to use an equation after high school”

David Hubel

What has theoretical component bought to neuroscience?

- The brain is squishy and messy. It may be impossible to understand the brain from a small number of (say 3?) principles just like in physics.
- There are too many facts! And almost every rule has an exception.
- However, most interesting facts are not singular. Instead, they must and can be connected to achieve a deeper understanding. And to build such a connection requires an integral approach, and mathematics are crucial during this process.

“...Equations force a model to be precise, complete, and self-consistent, and they allow its full implications to be worked out. It is not difficult to find word models in the conclusions sections of older neuroscience papers that sound reasonable but, when expressed as mathematical models, turn out to be inconsistent and unworkable. Mathematical formulation of a model forces it to be self-consistent and, although self-consistency is not necessarily truth, self-inconsistency is certainly falsehood.”

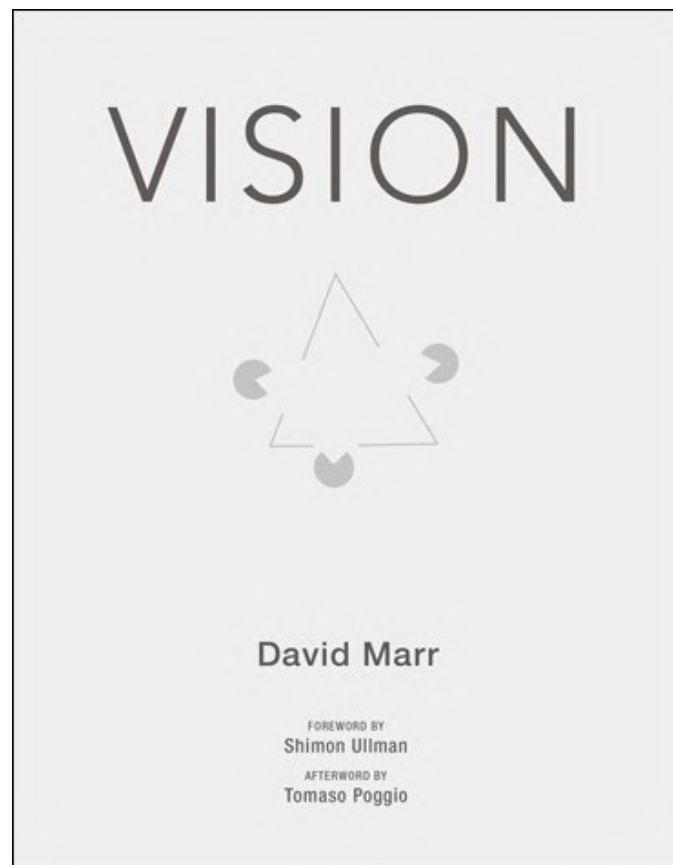
Larry Abbott
Theoretical Neuroscience is rising
Neuron 2008



David Marr
1945-1980



Henry Markram
1962 -

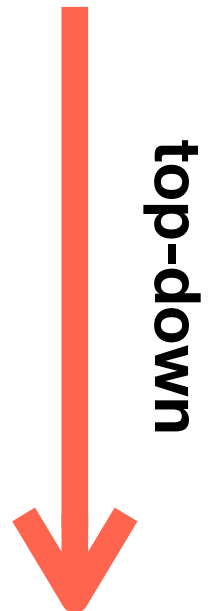


blue brain project



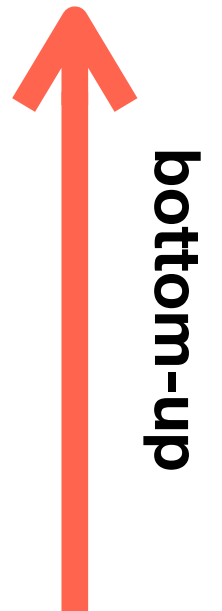
David Marr's three level theory

- **Computational level:** Identify the computational problem and task that the brain solves.
- **Algorithmic level:** Find the mathematical procedures that solve the problem.
- **Implementation level:** How the algorithms are realized by the nervous system.



Henry Markram's three level theory

- **Systems level:** describe how population neural dynamics and behaviors emerge from ensembles of neurons.
- **Cellular level:** develop biophysically accurate models to describe input-output relationships of different cell types.
- **Structure level:** identify how neurons are statistically connected to each other in a circuit.



The structure of an information processing machine does not tell you what computation it performs

Anthony Movshon

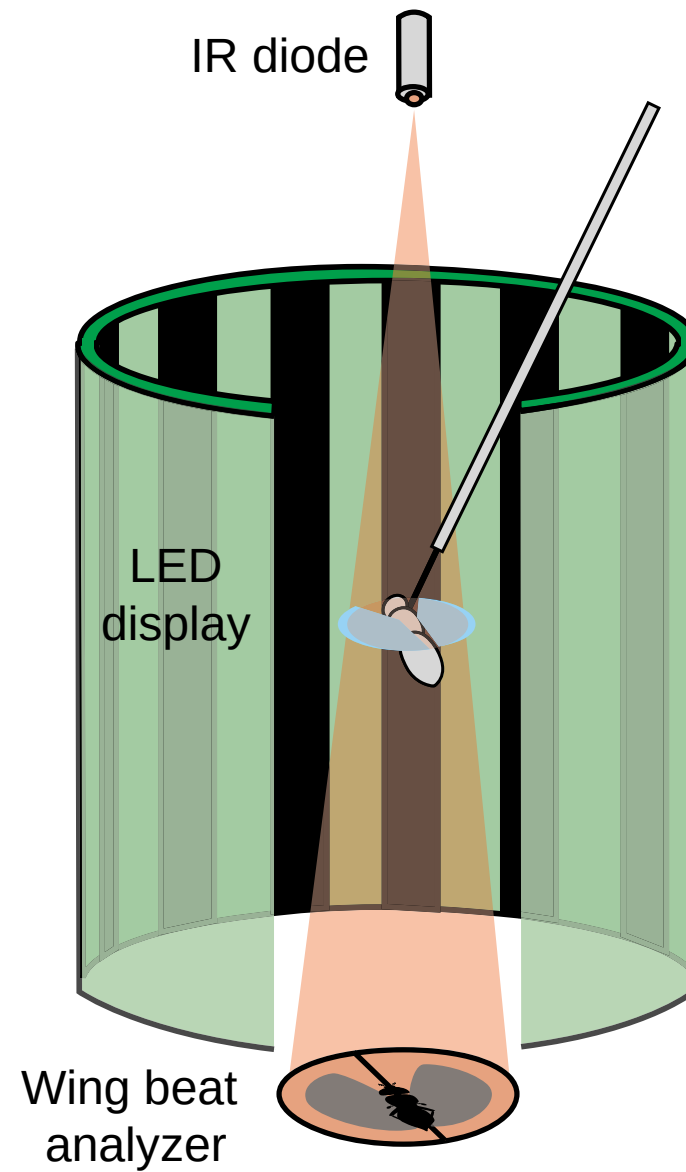
Circuit diagrams or connectomes are *absolutely* necessary but *completely* insufficient for understanding nervous system function.

Eve Marder

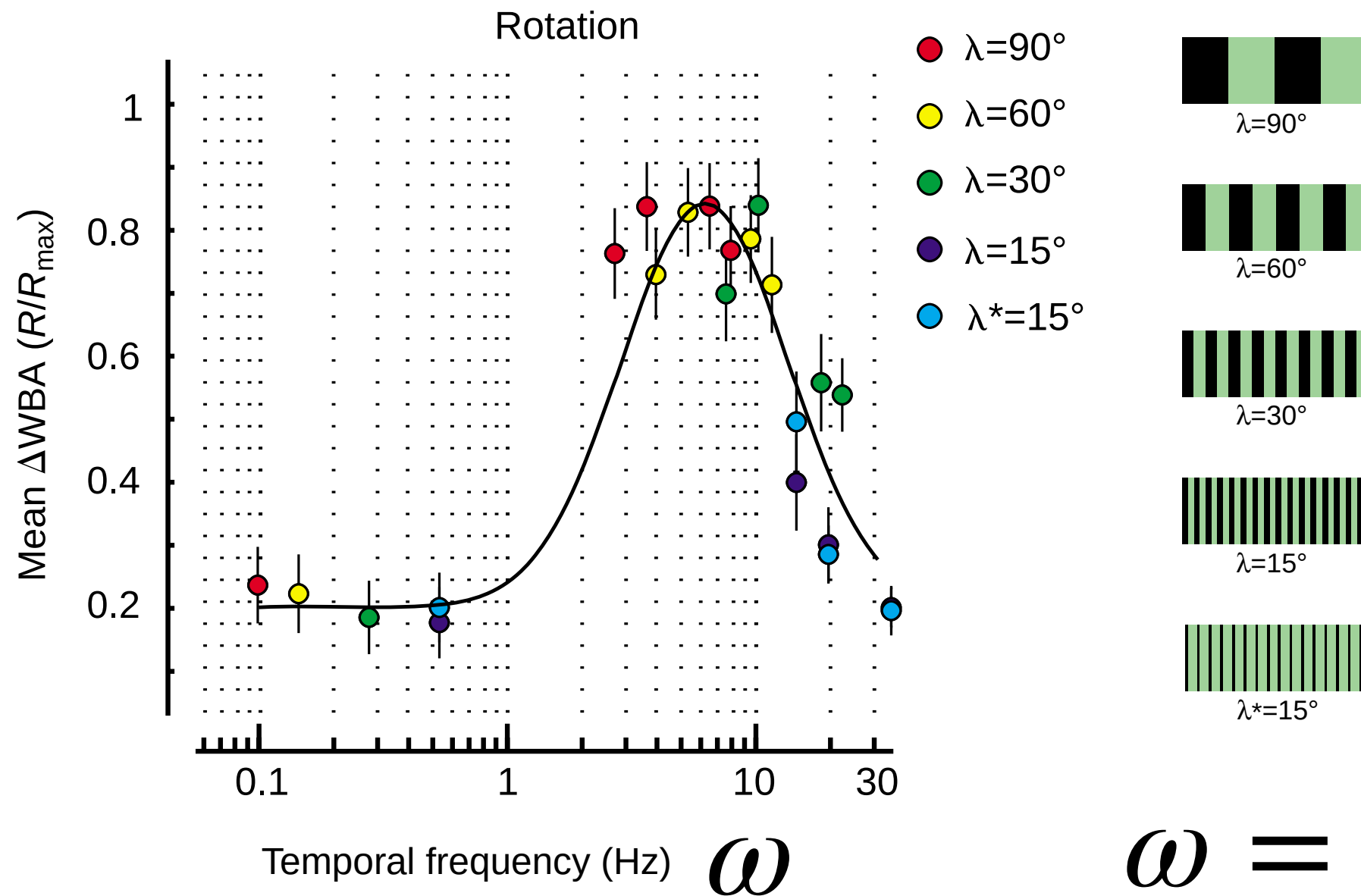
Motion detection, an example



Optomotor response in fly

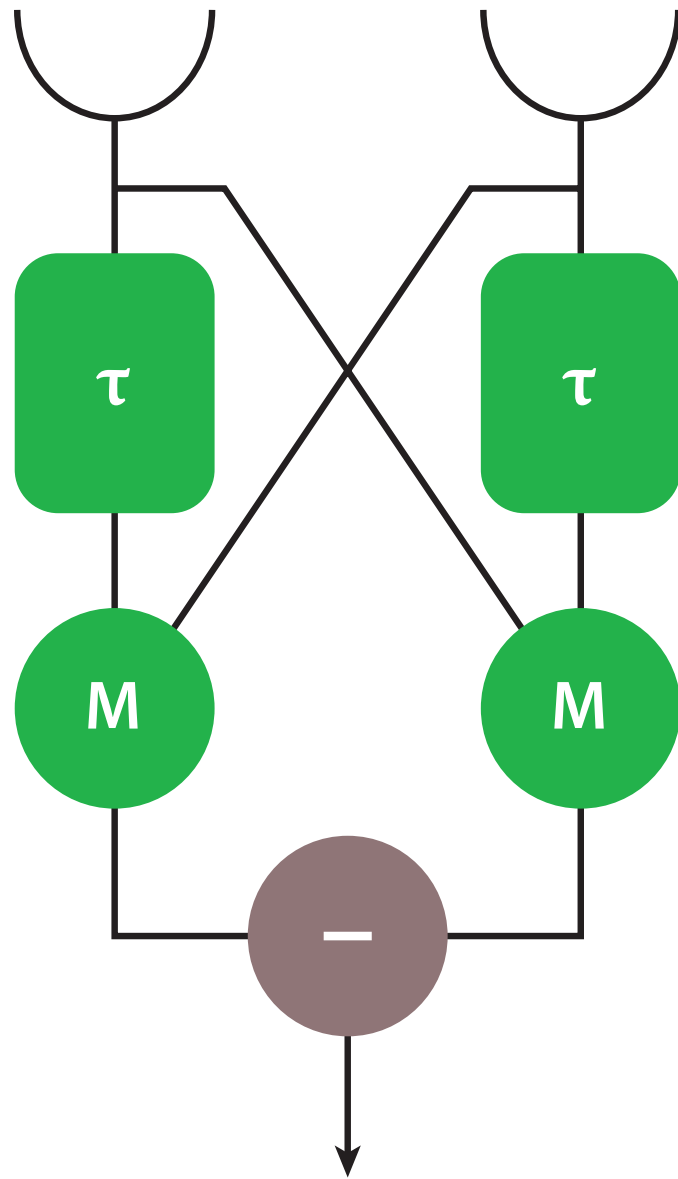


Optomotor response in fly

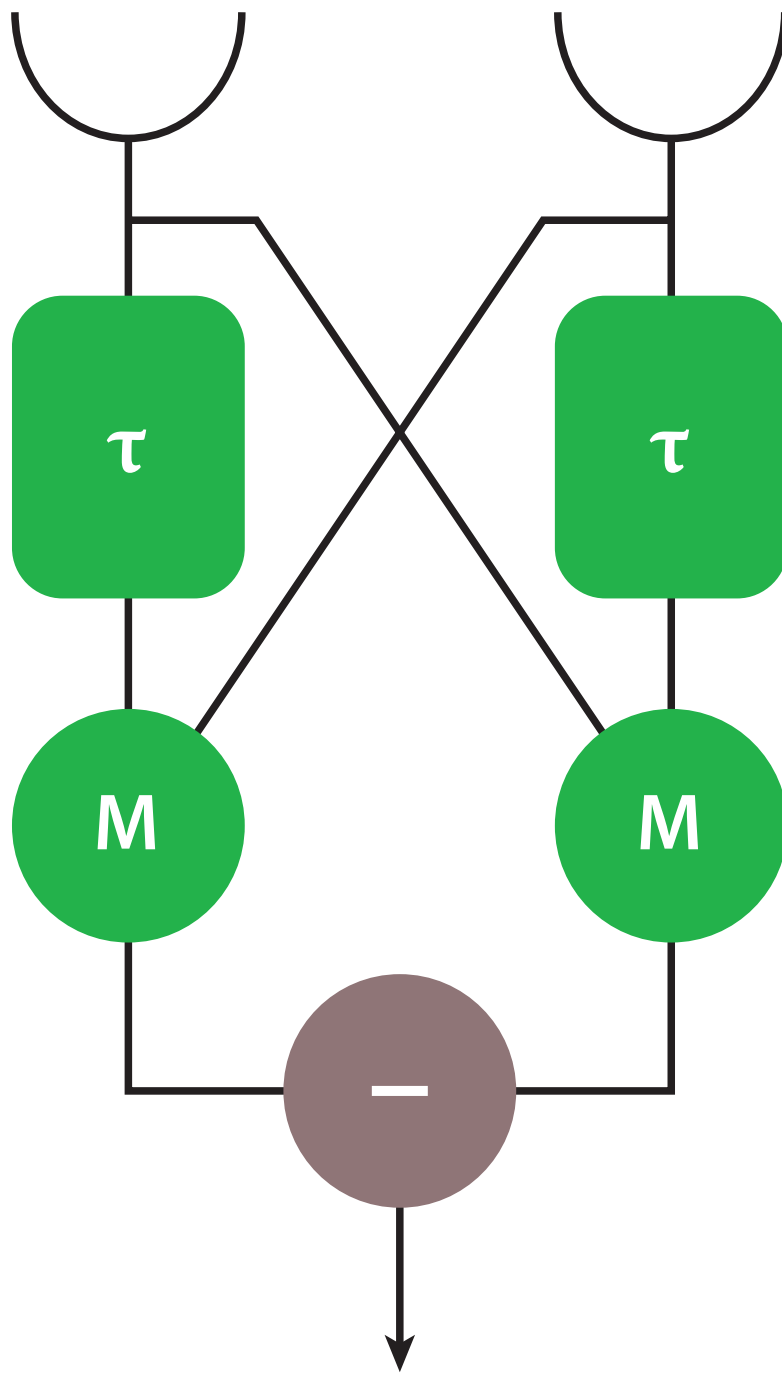
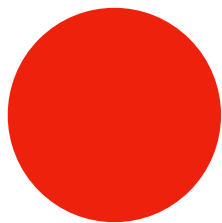


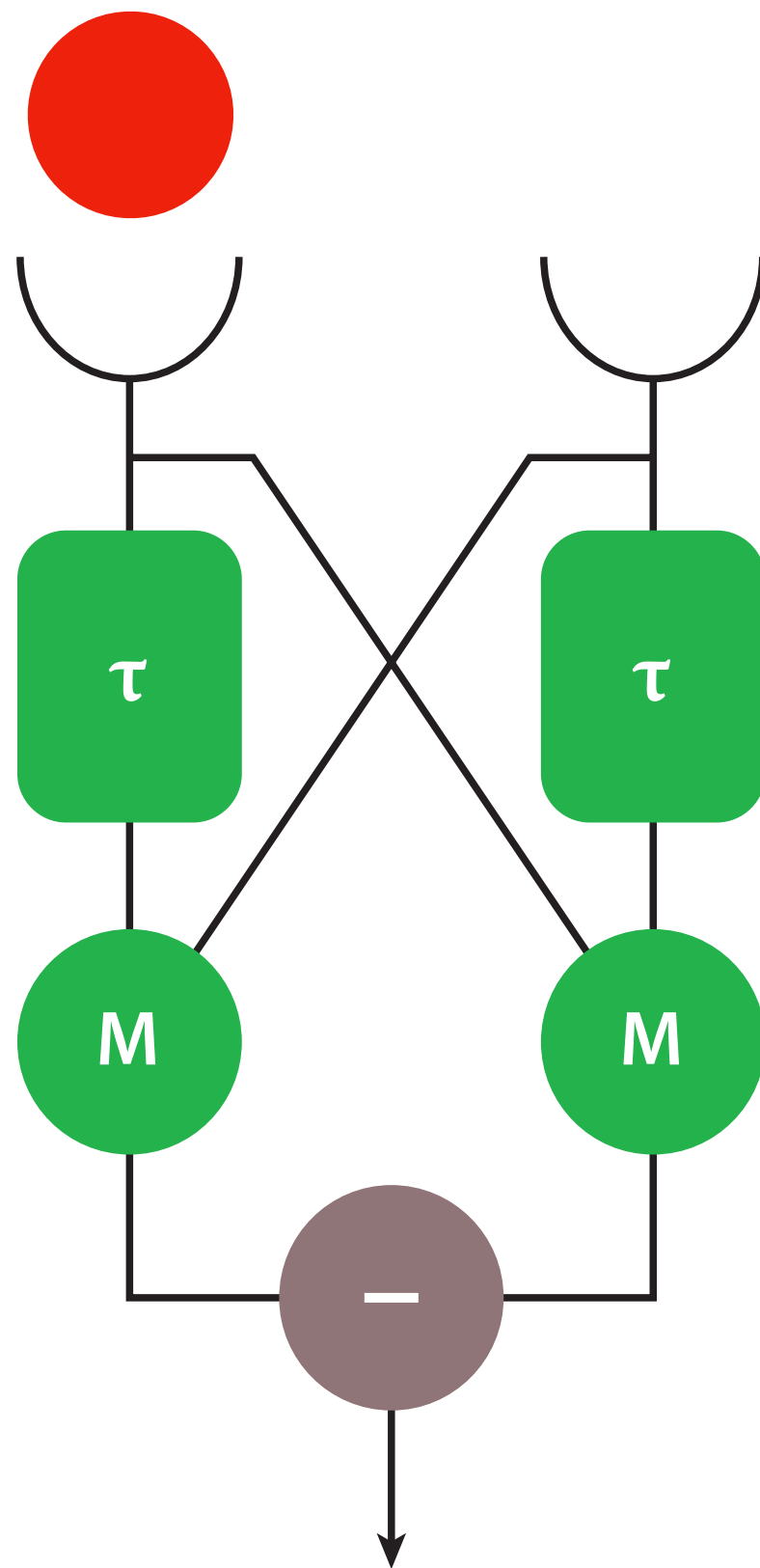
$$\omega = 2\pi\nu/\lambda$$

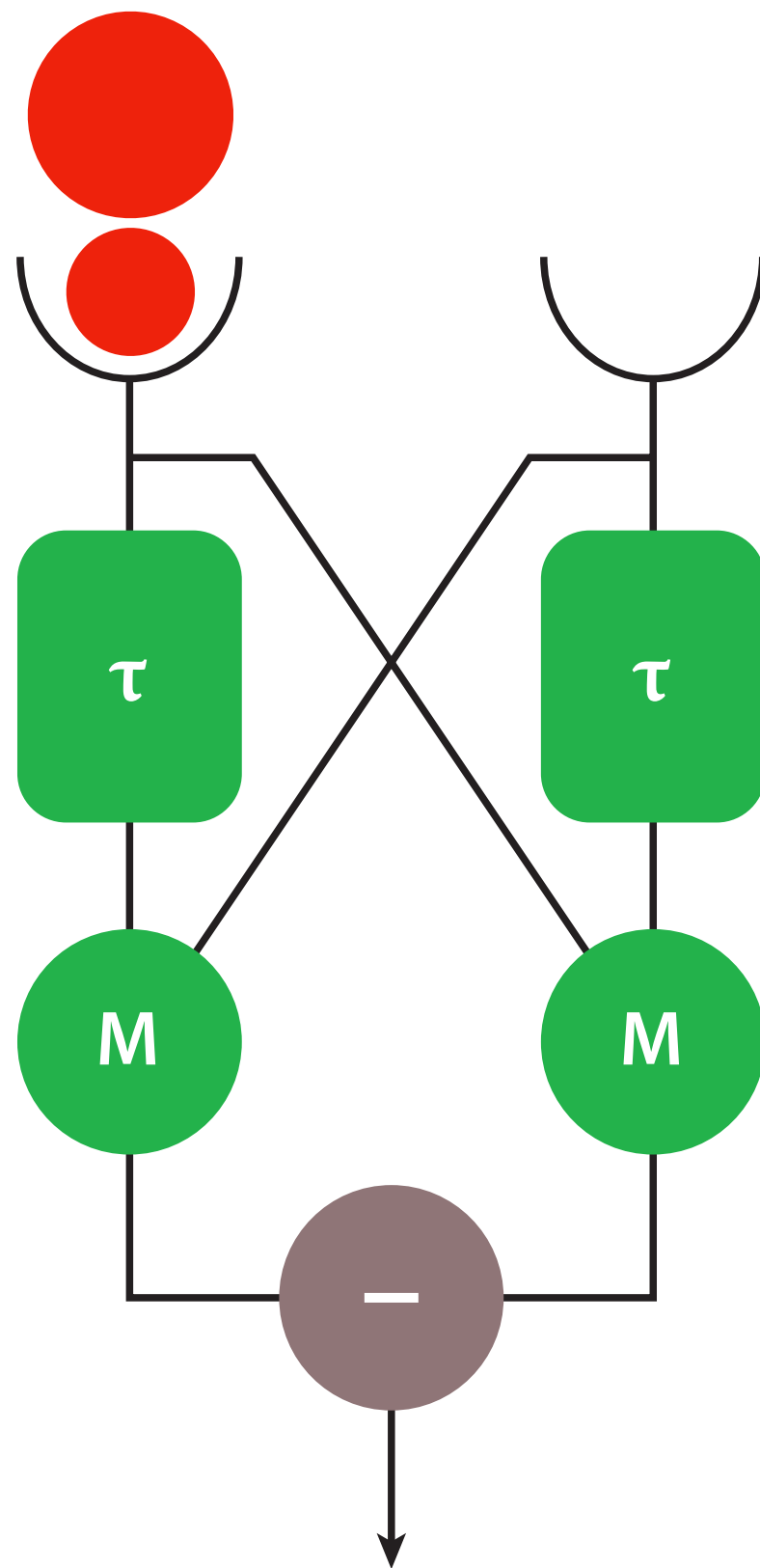
Hassenstein-Reichardt Detector Model

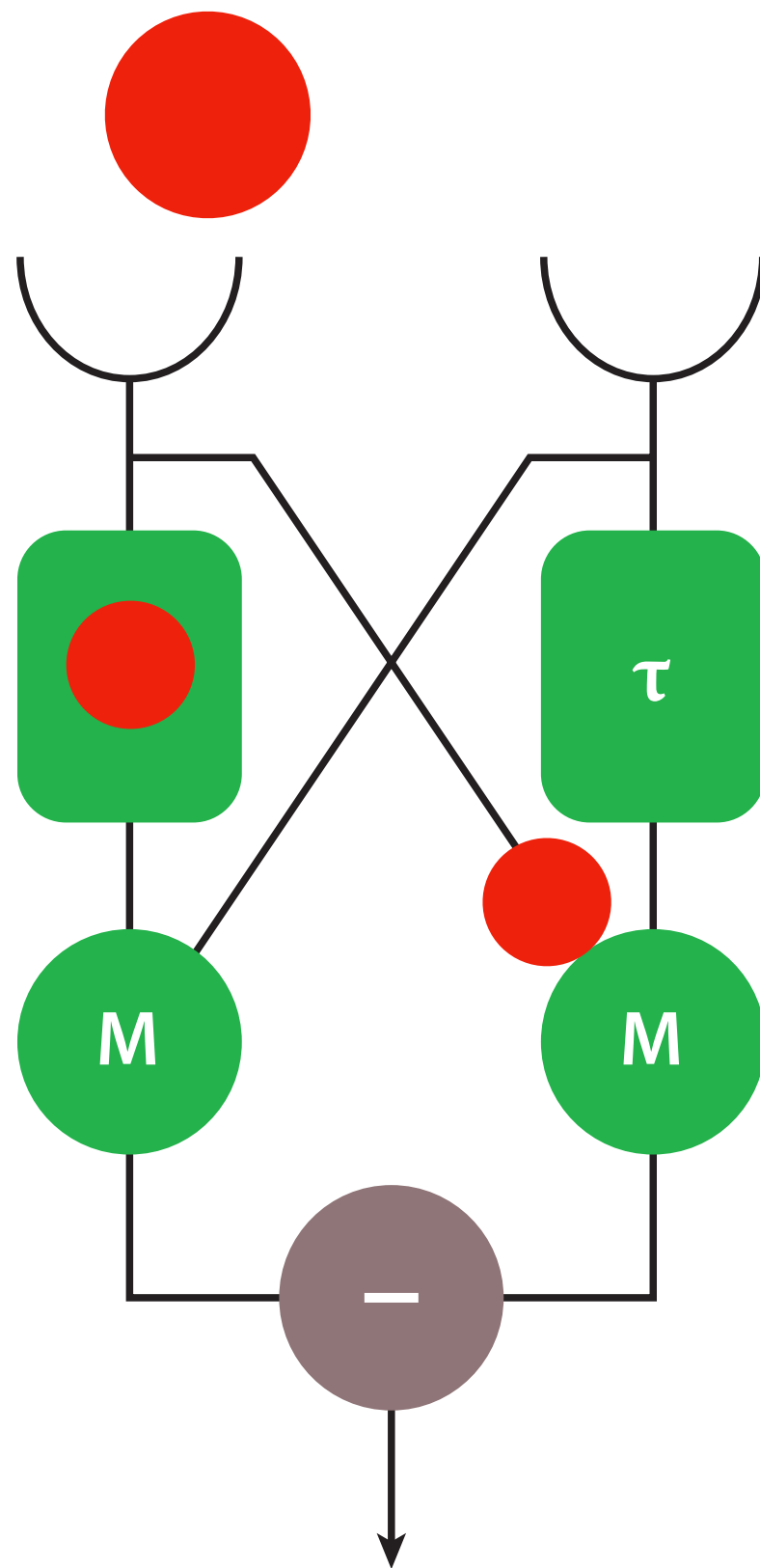


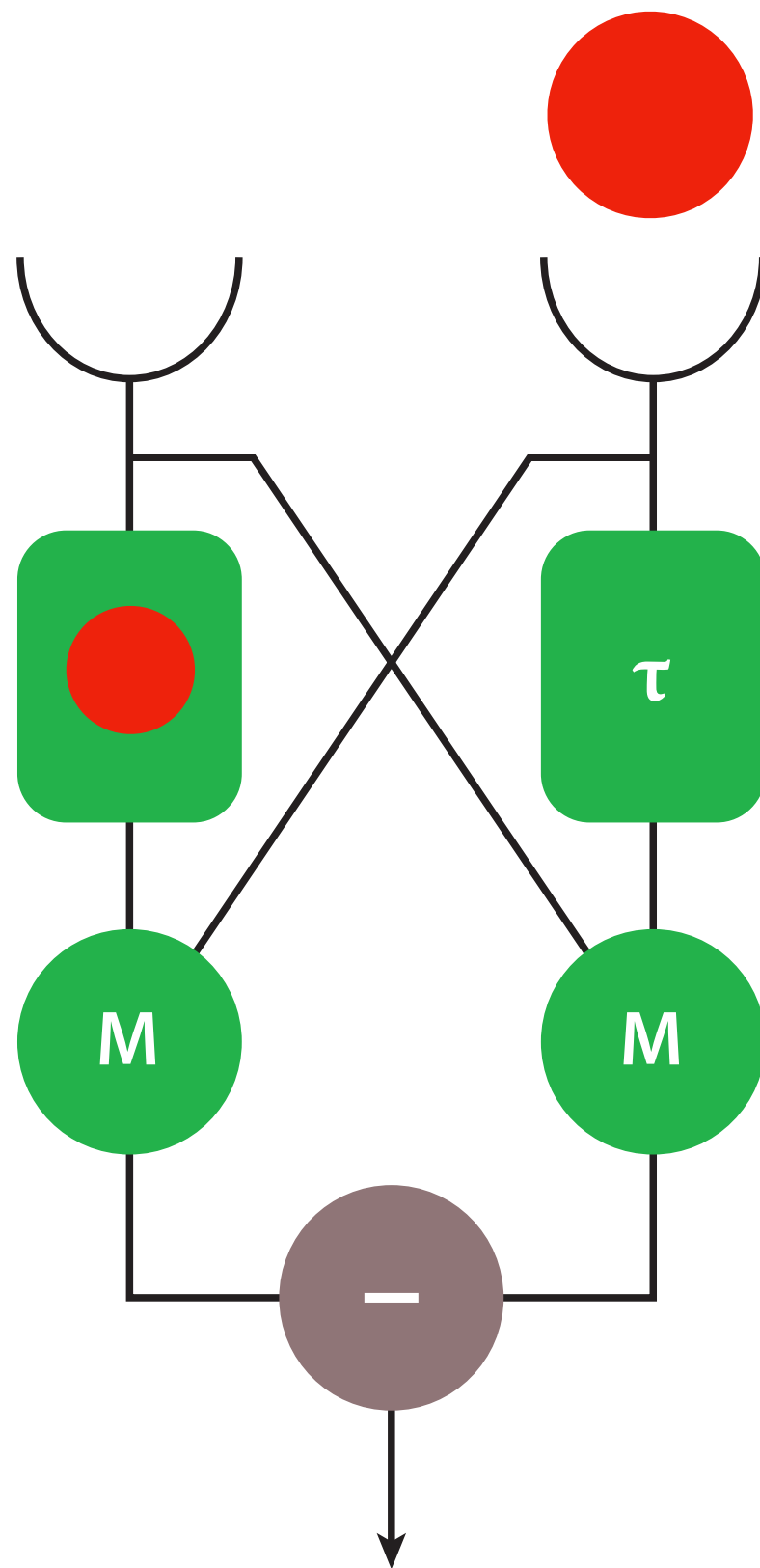
Werner Reichardt

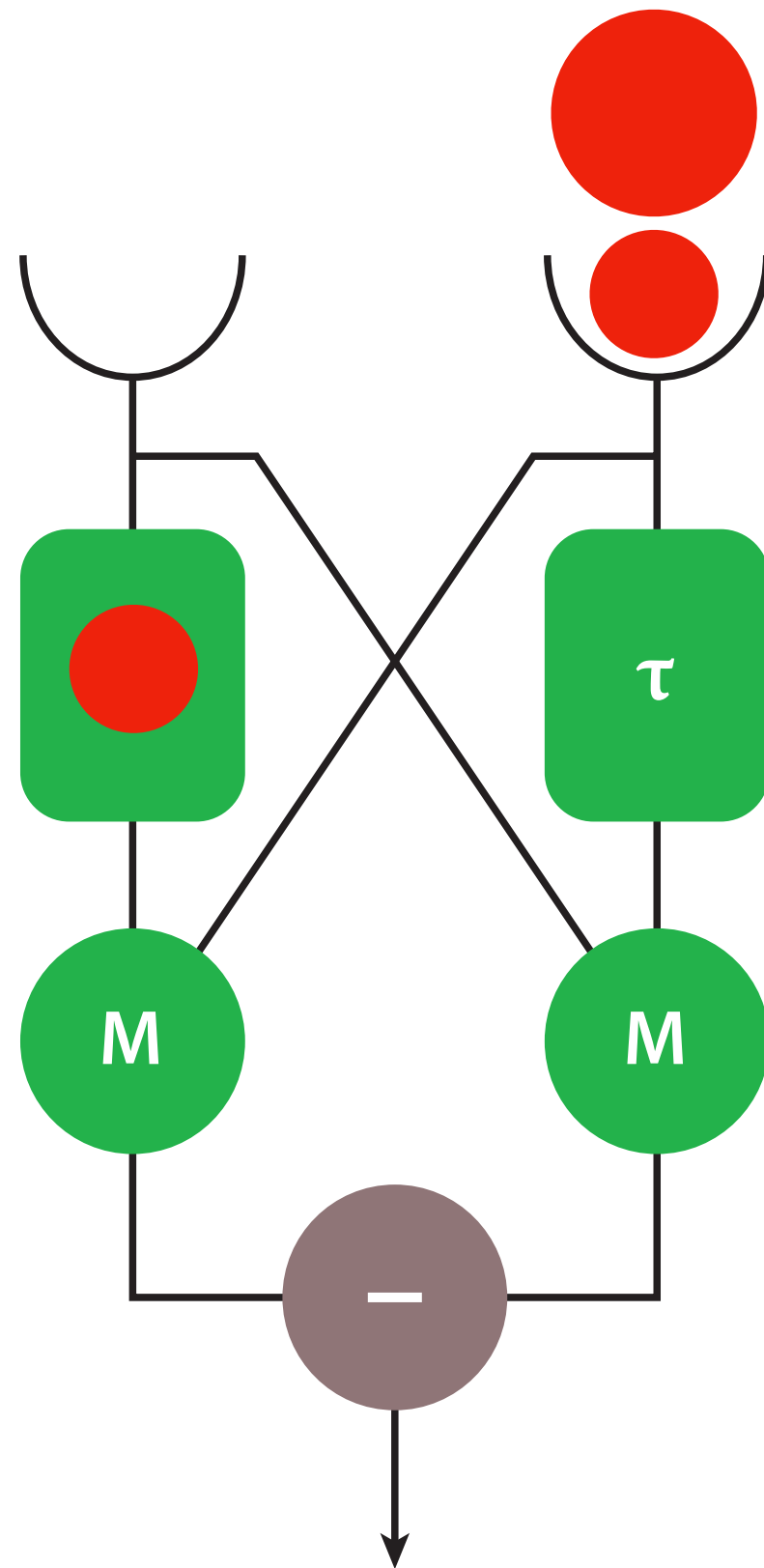


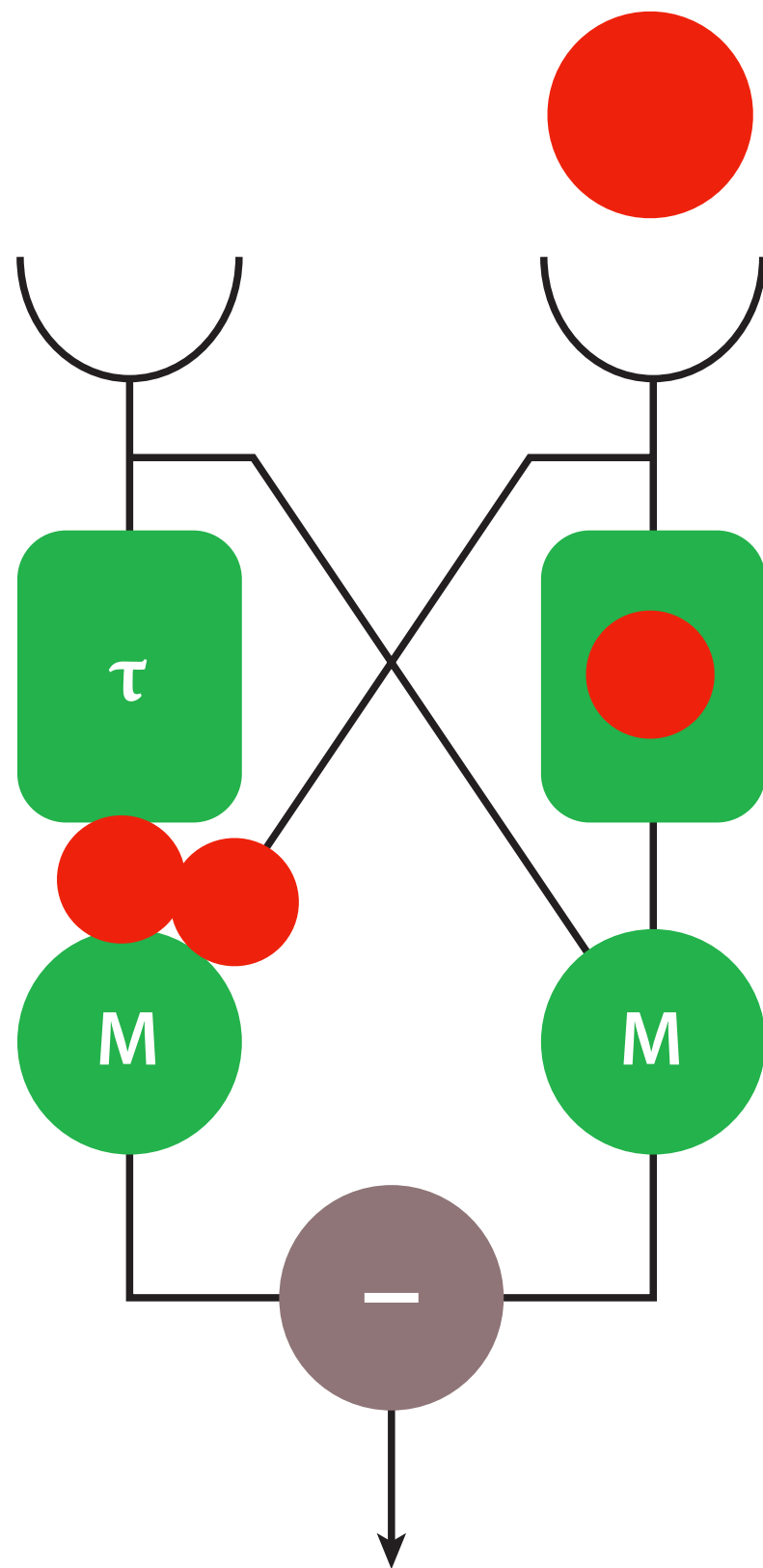


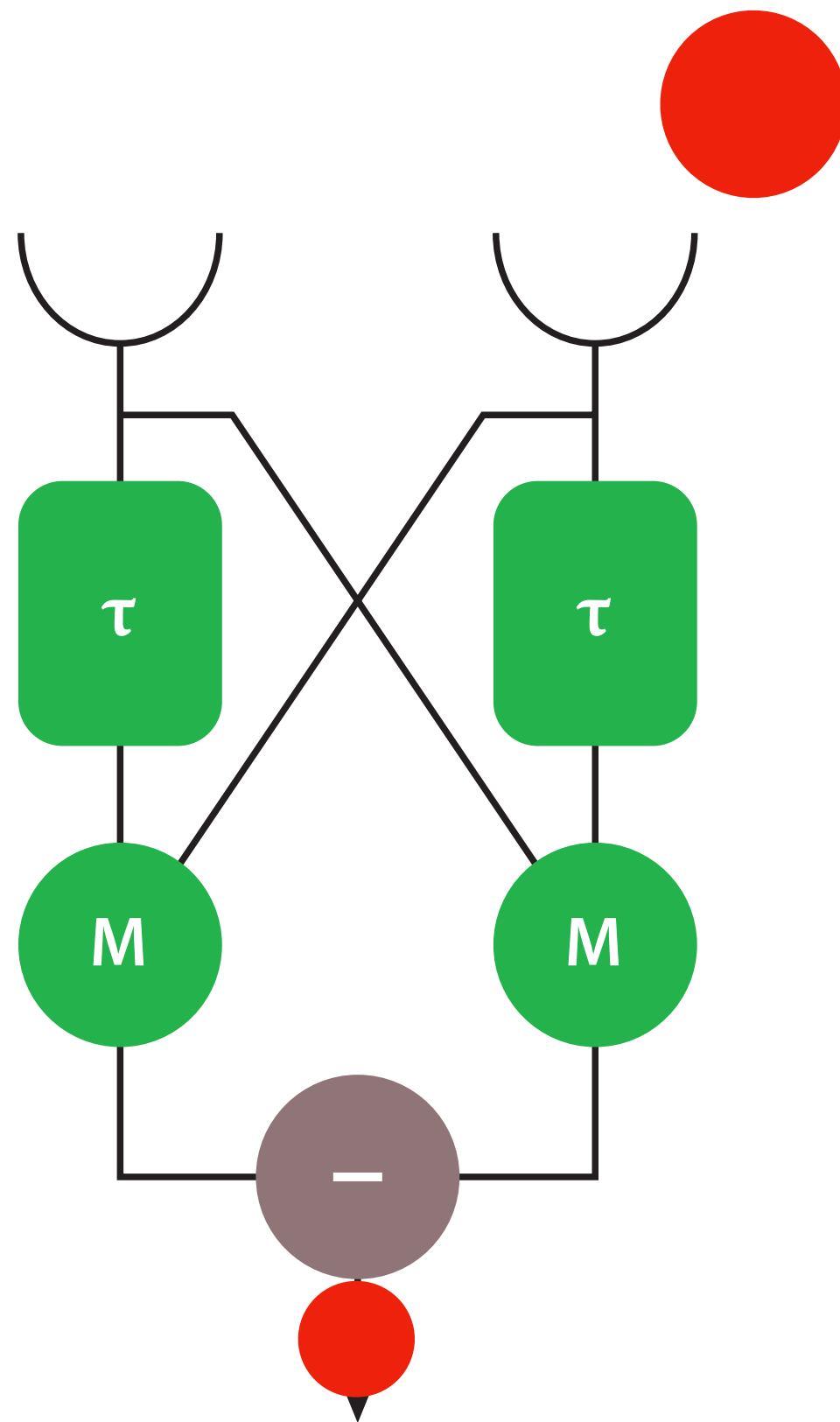




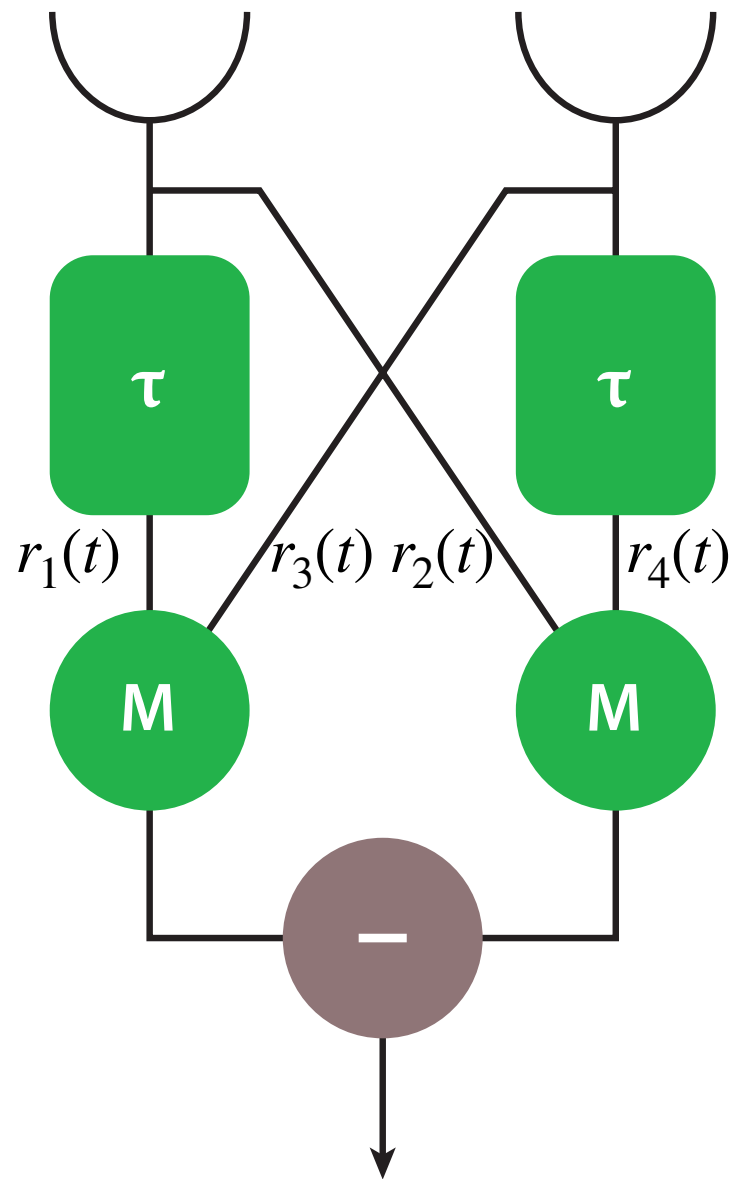








Hassenstein-Reichardt Detector Model



$$r_1(t) = \int_0^\infty s_1(t - \tau) D(\tau) d\tau; \quad r_2(t) = \int_0^\infty s_1(t - \tau) \delta(\tau) d\tau$$

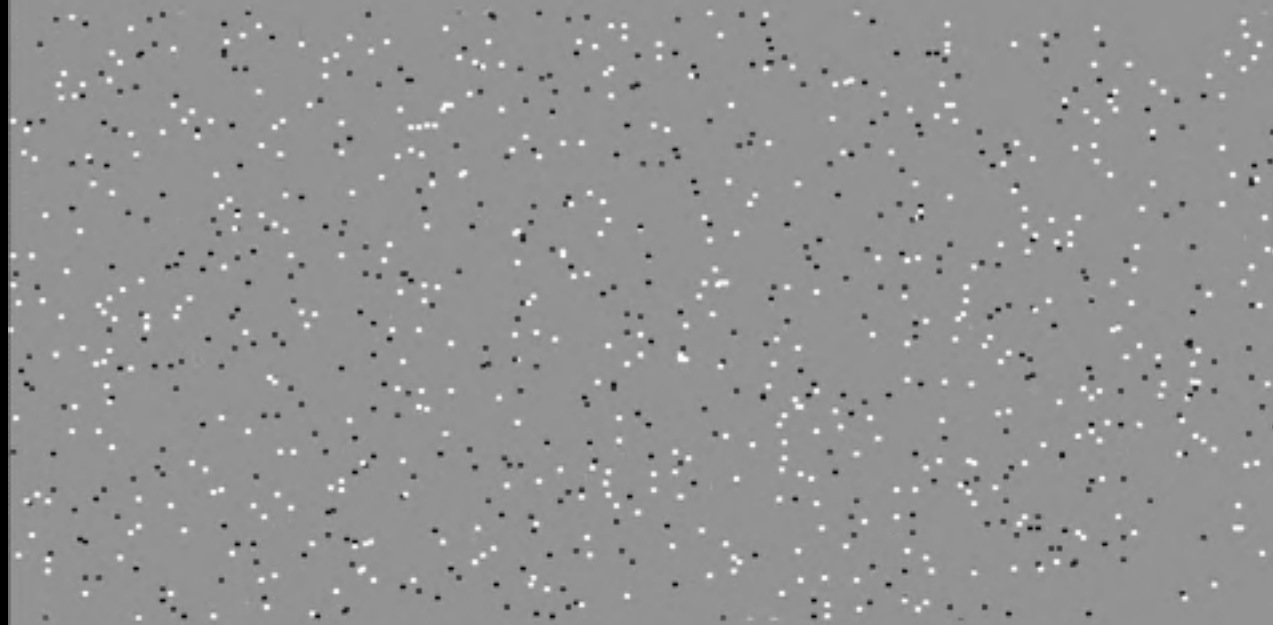
$$r_3(t) = \int_0^\infty s_2(t - \tau) \delta(\tau) d\tau; \quad r_4(t) = \int_0^\infty s_2(t - \tau) D(\tau) d\tau;$$

$$R(t) = r_1(t)r_3(t) - r_2(t)r_4(t)$$

$$D(\tau) = \frac{1}{\tau_0} \exp(-\tau/\tau_0)$$

$$\langle R \rangle = \frac{\omega \tau_0}{\omega^2 \tau_0^2 + 1}$$

Phi: displacements to right

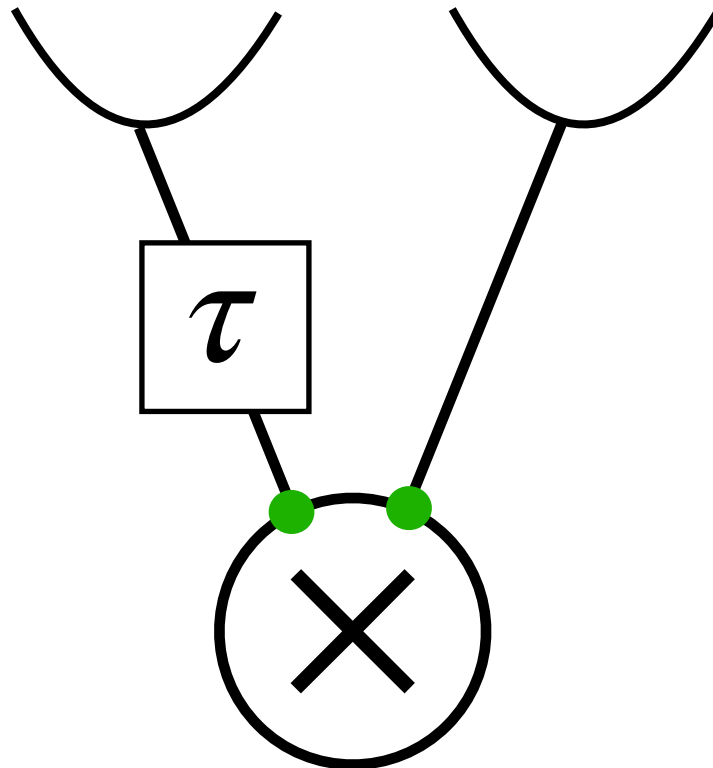


adapted from Damon Clark's slide

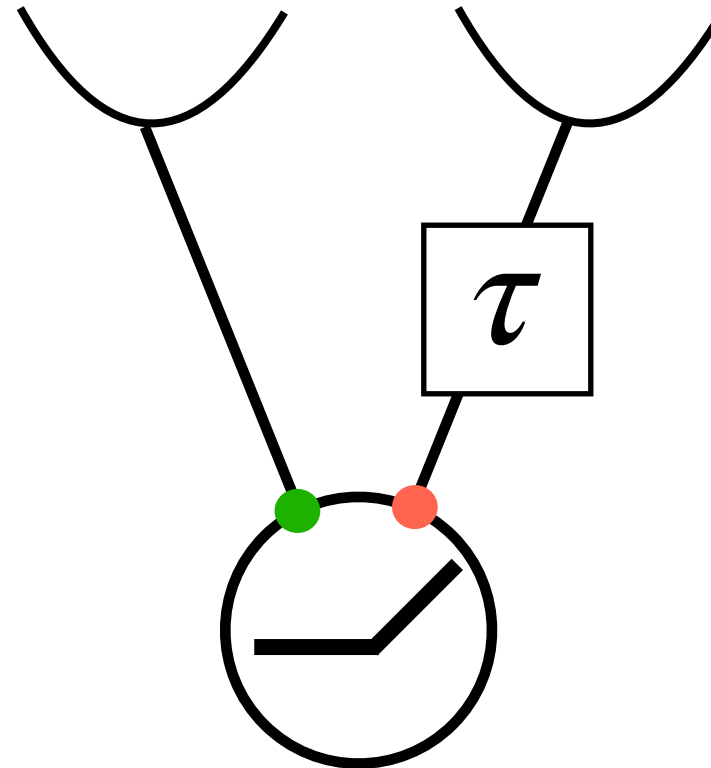
preferred
direction



null
direction

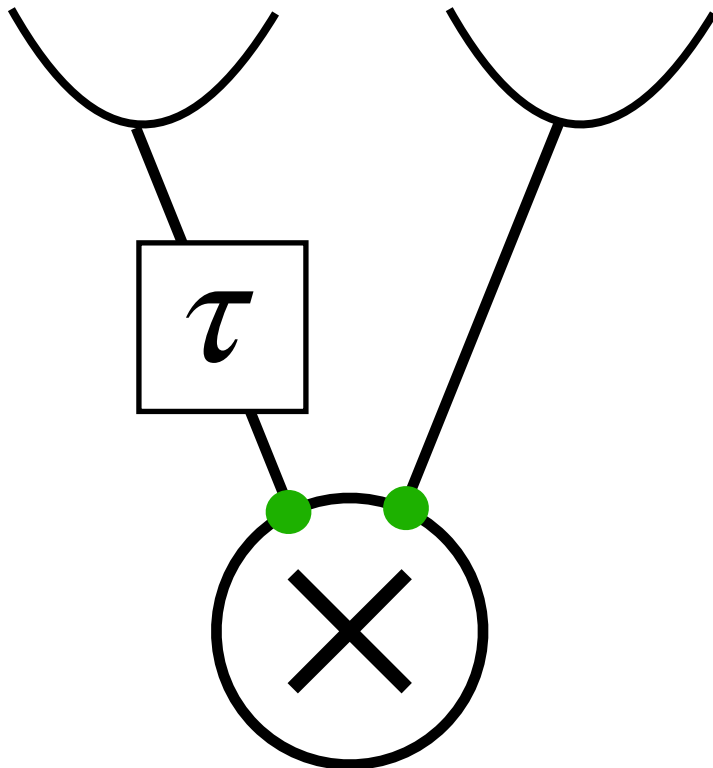
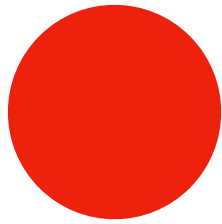


Reichardt model

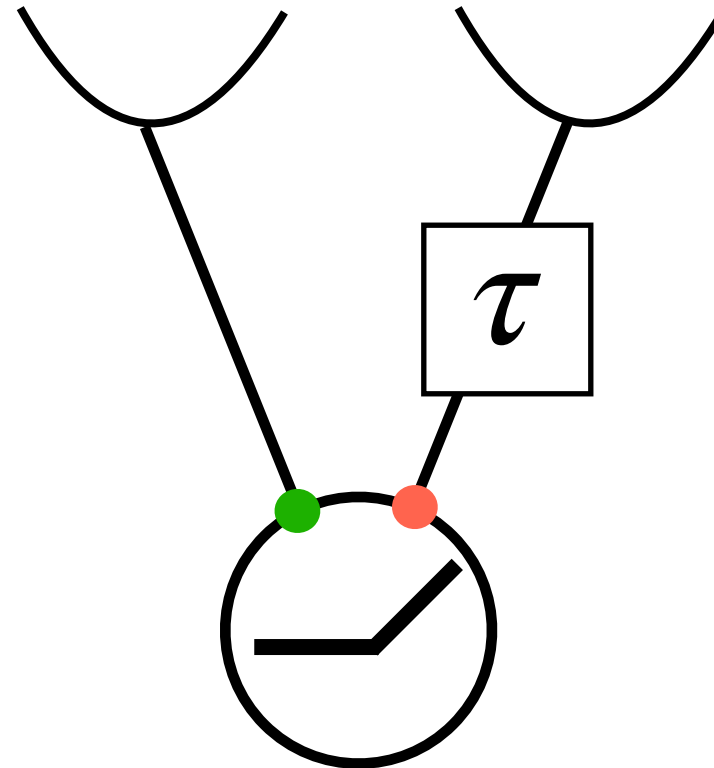
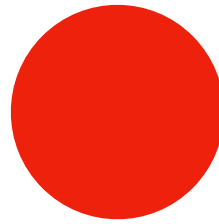


barlow-levick model

preferred
direction

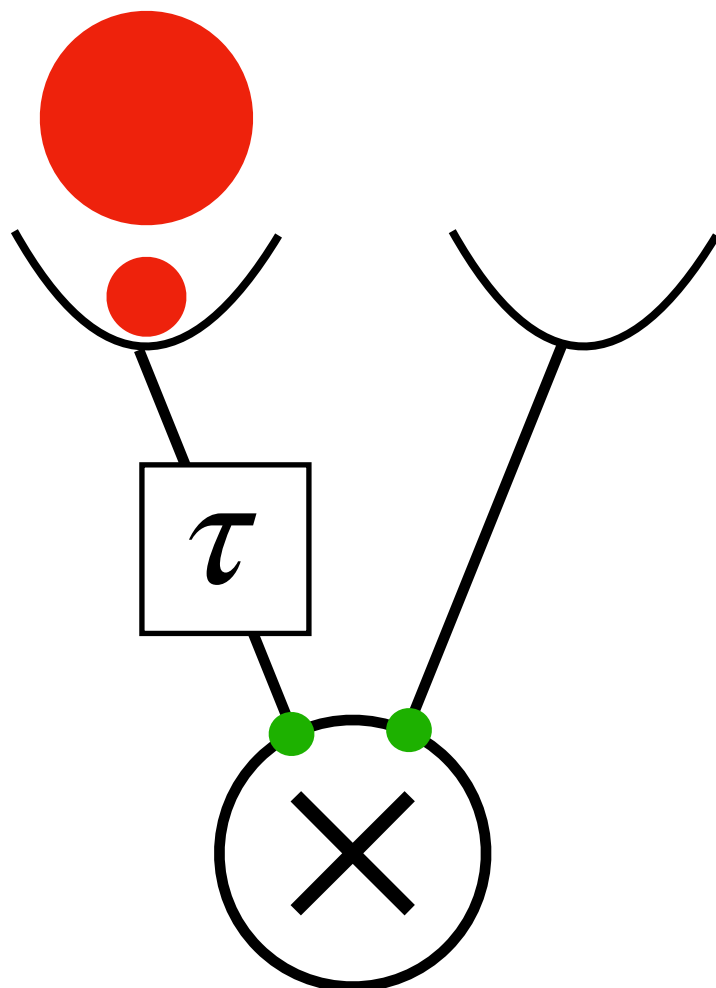


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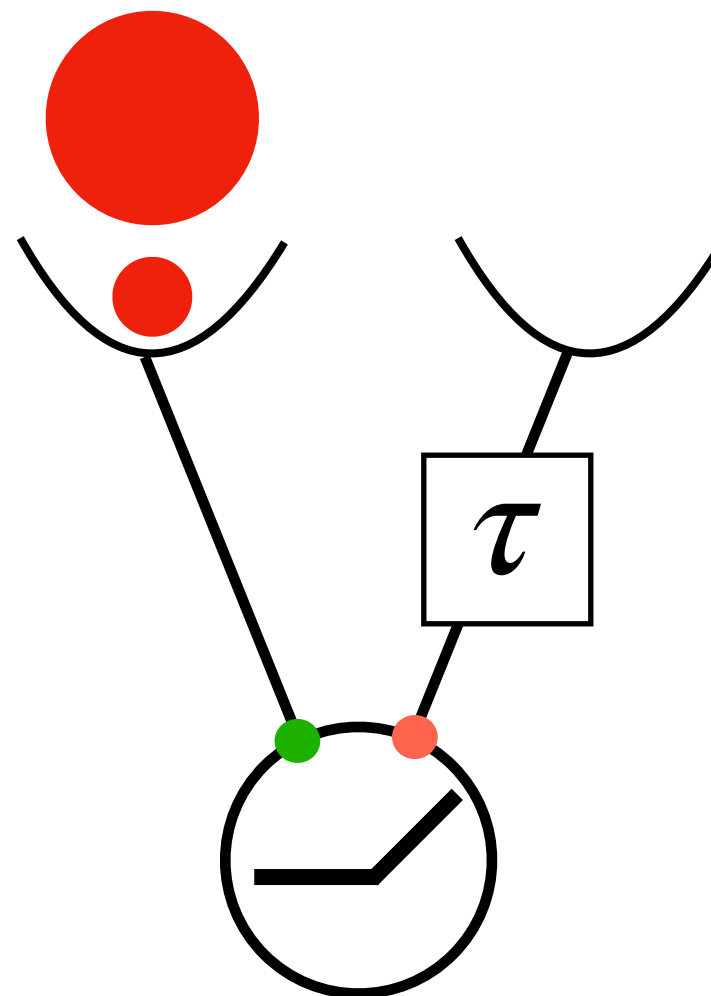


barlow-levick model

preferred
direction

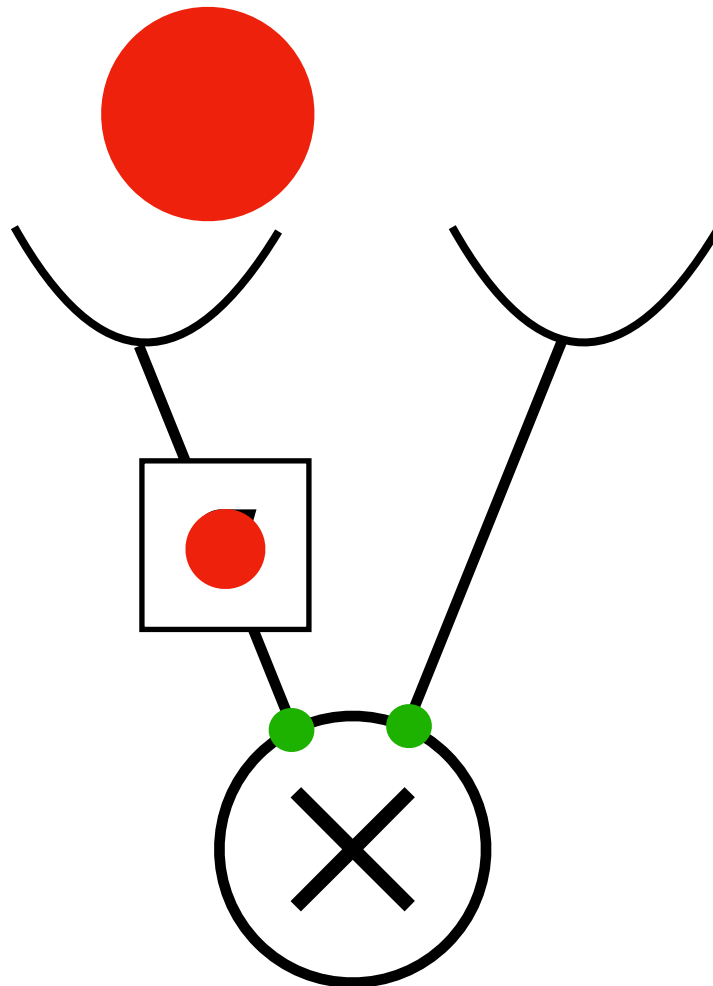


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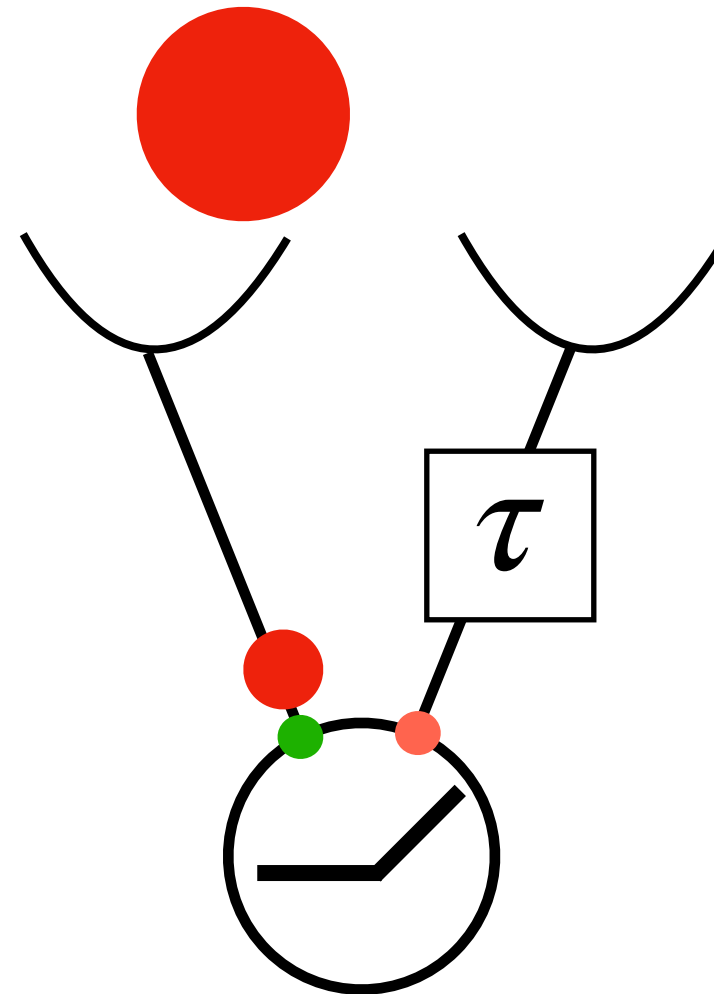


barlow-levick model

preferred
direction

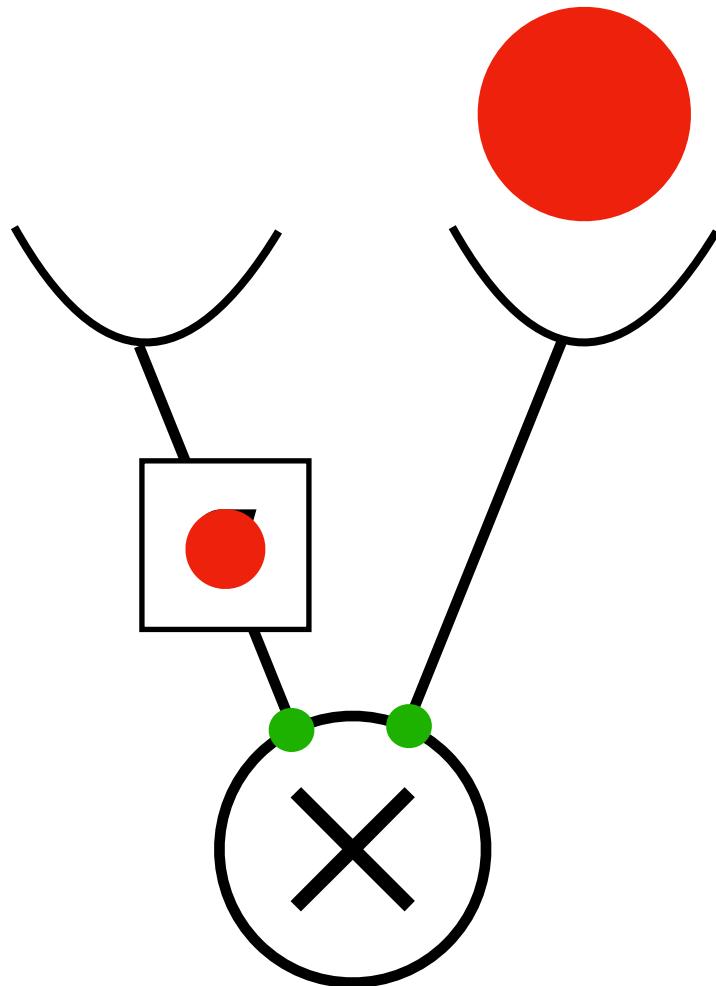


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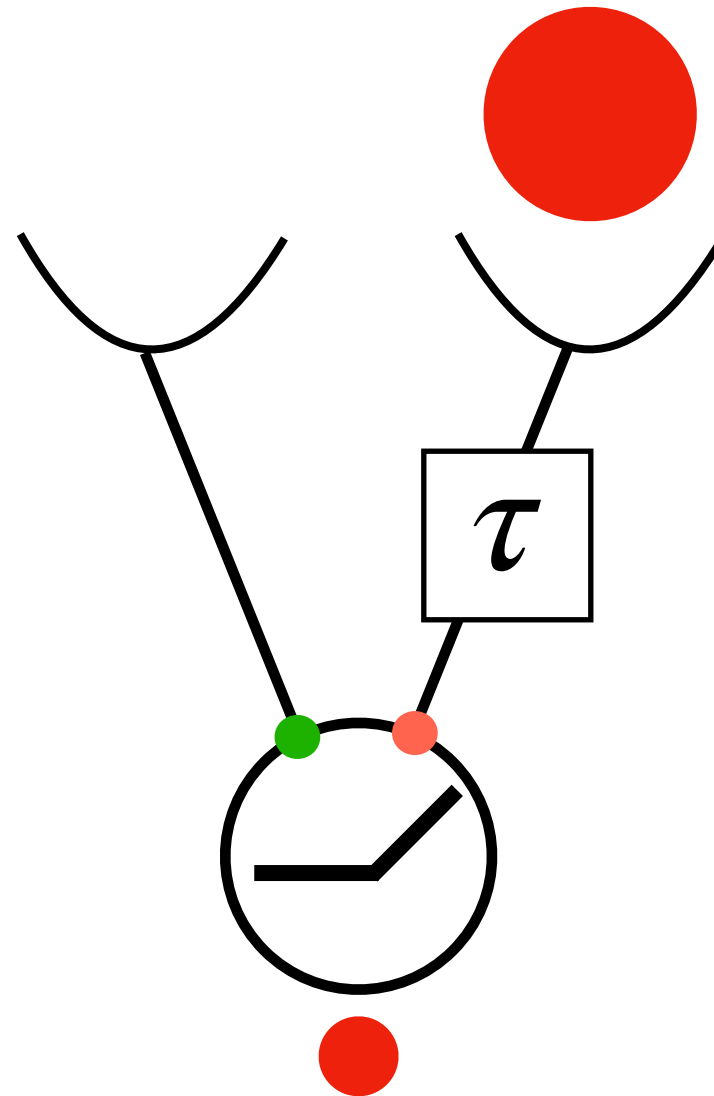


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direction

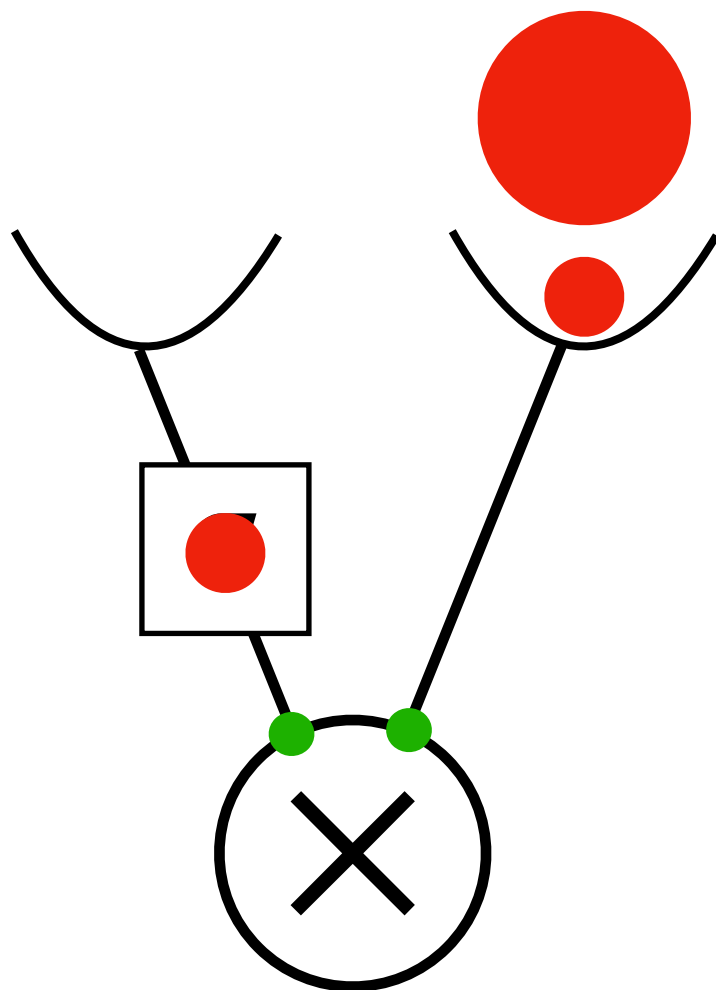


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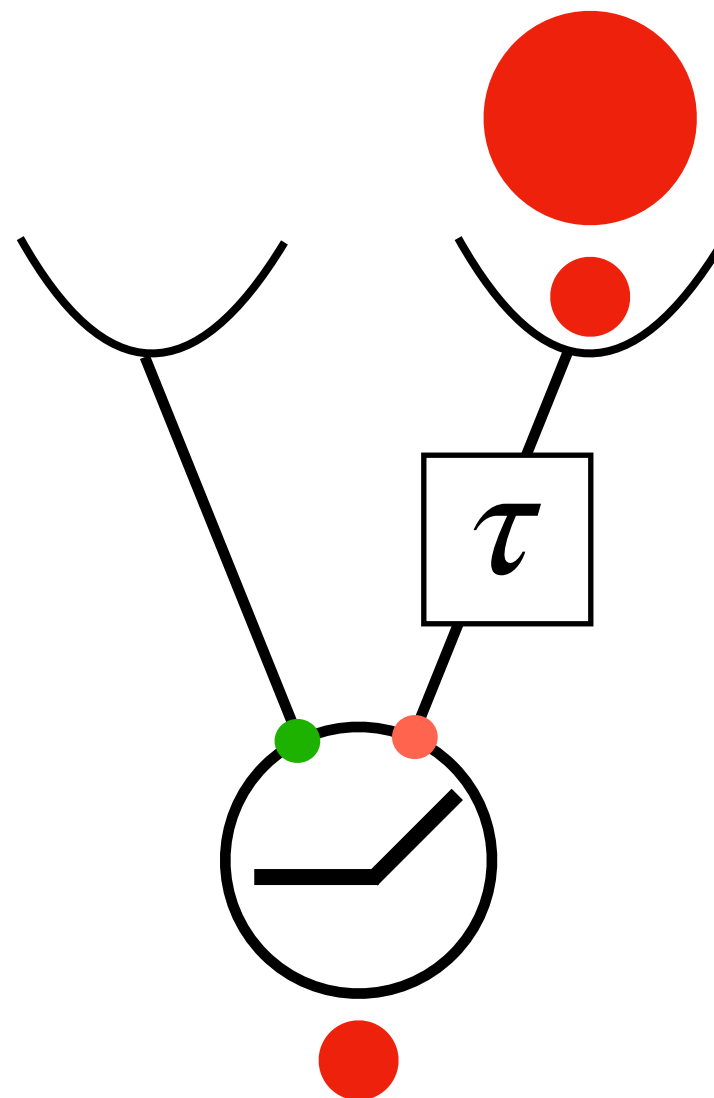


barlow-levick model

preferred
direction

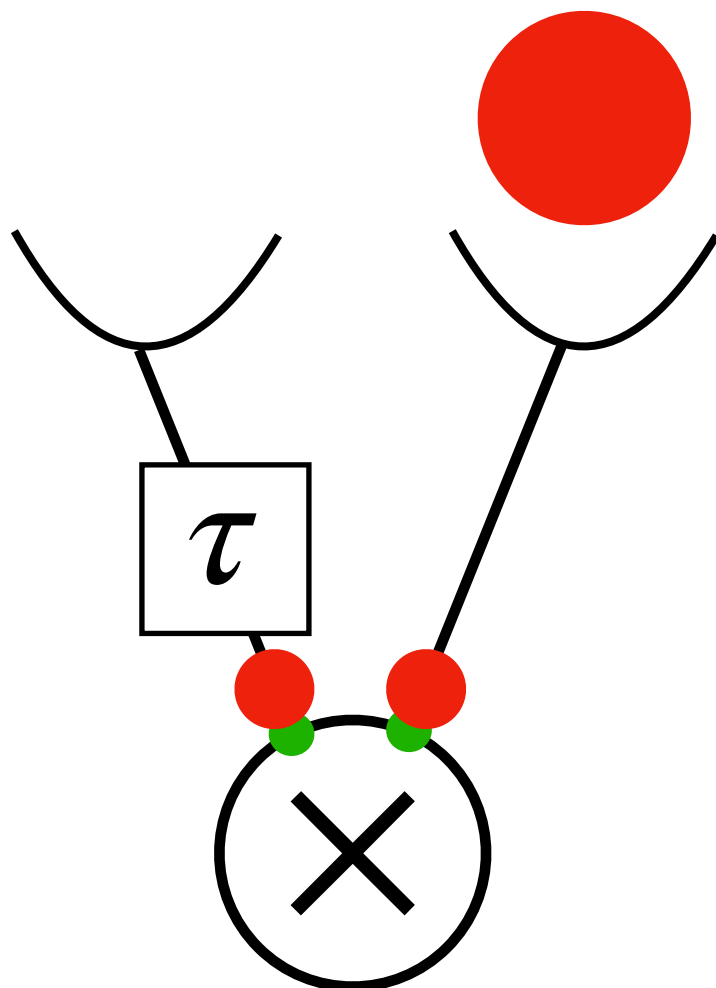


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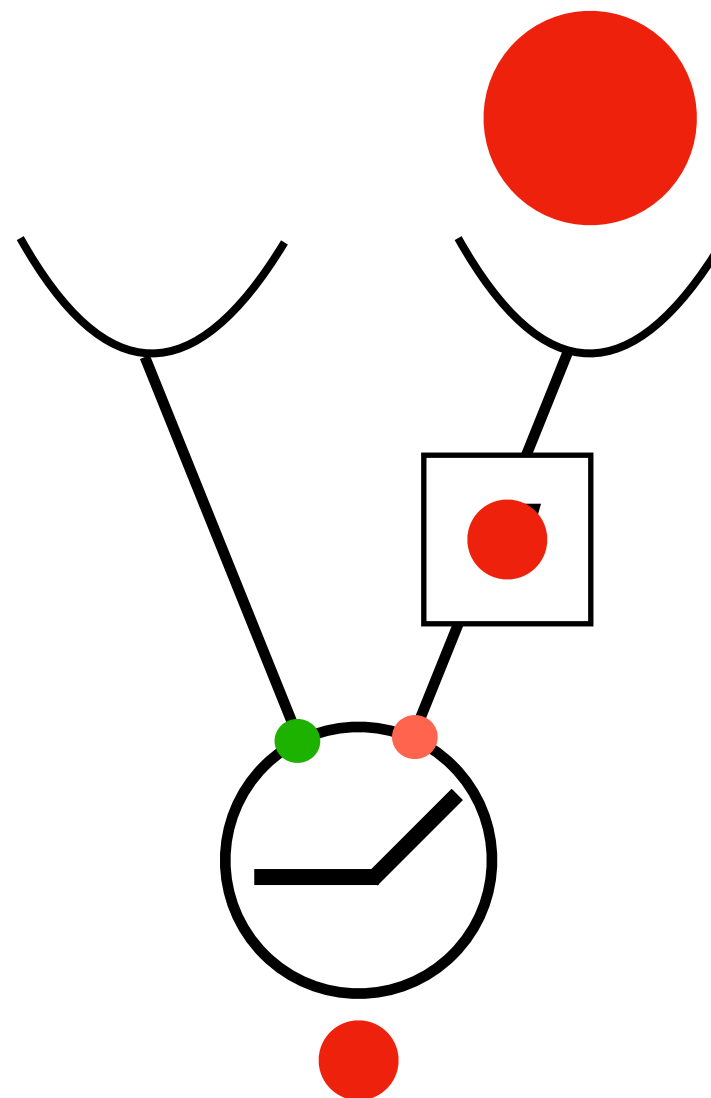


barlow-levick model

preferred
direction

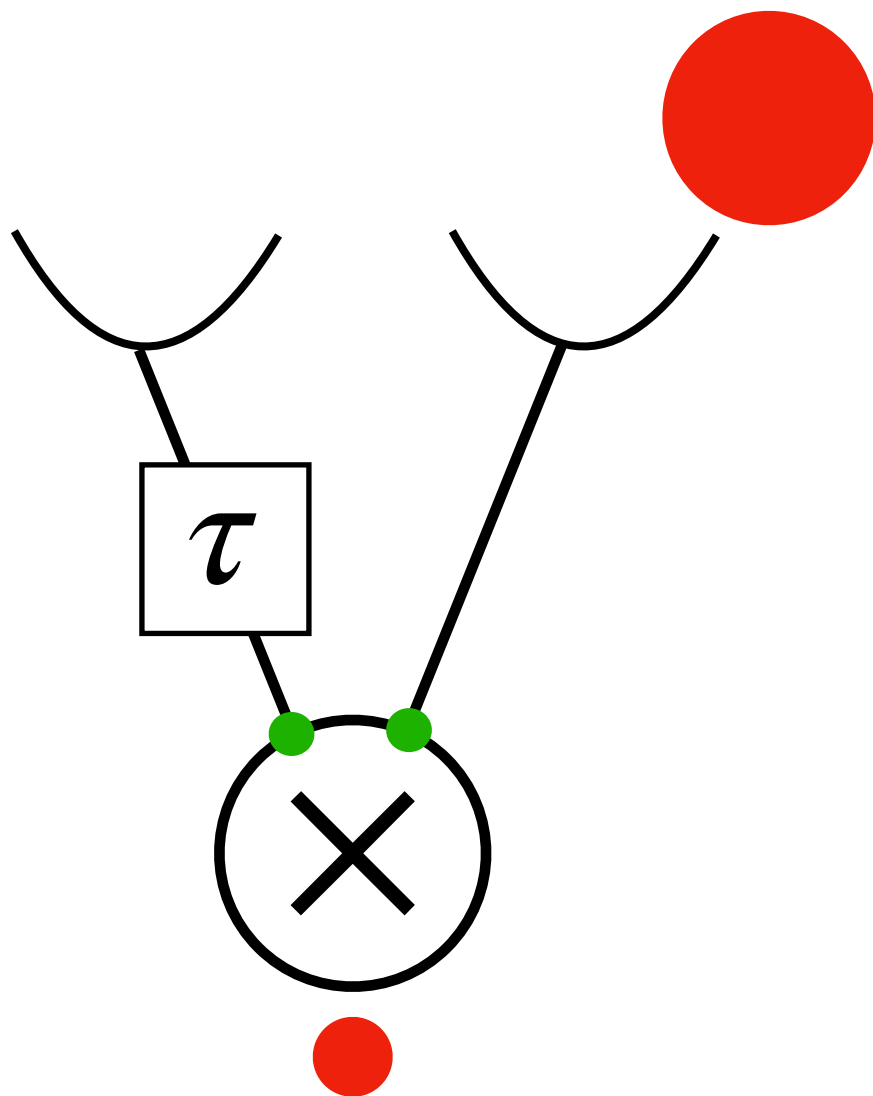


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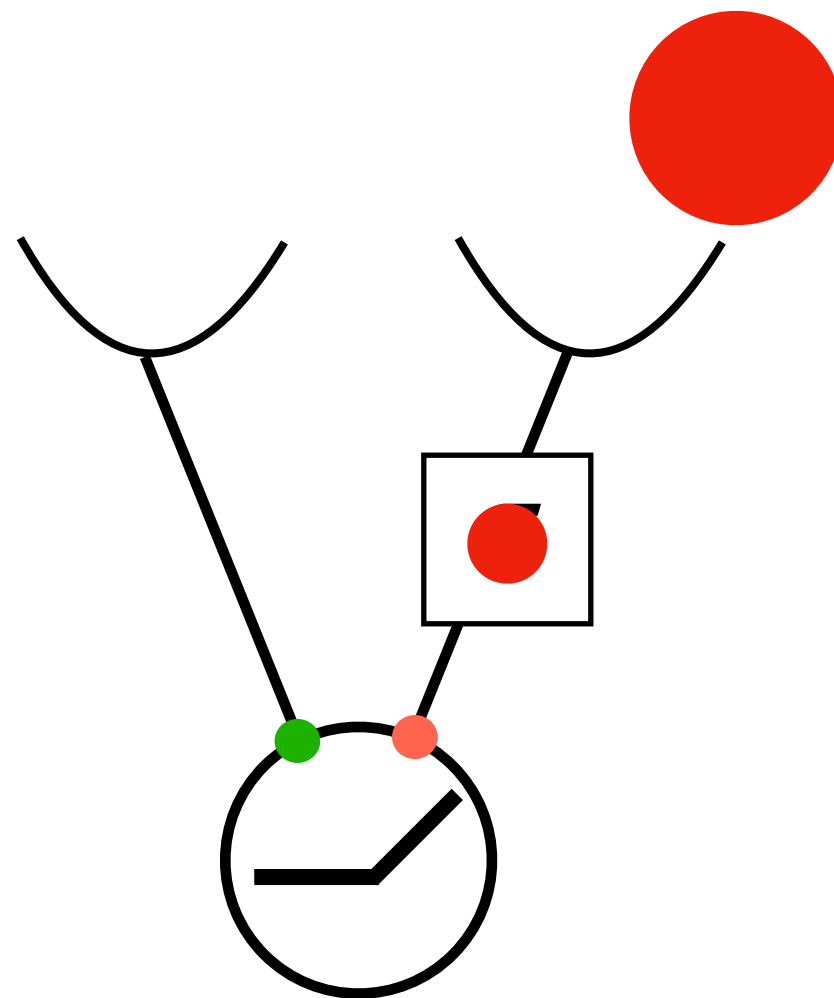


barlow-levick model

preferred
direction

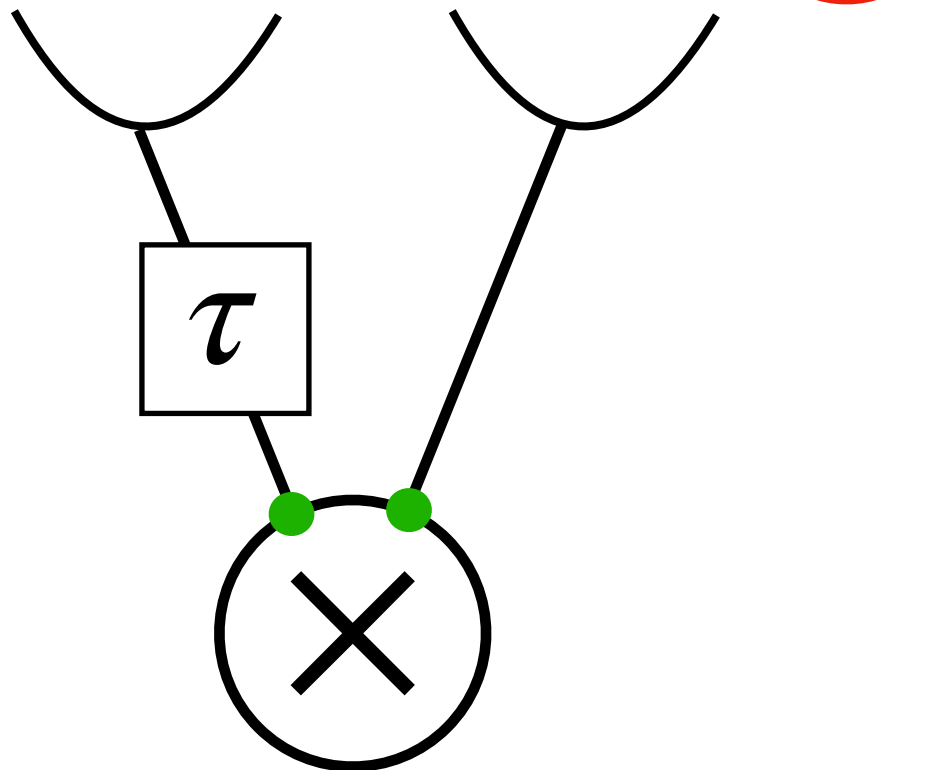


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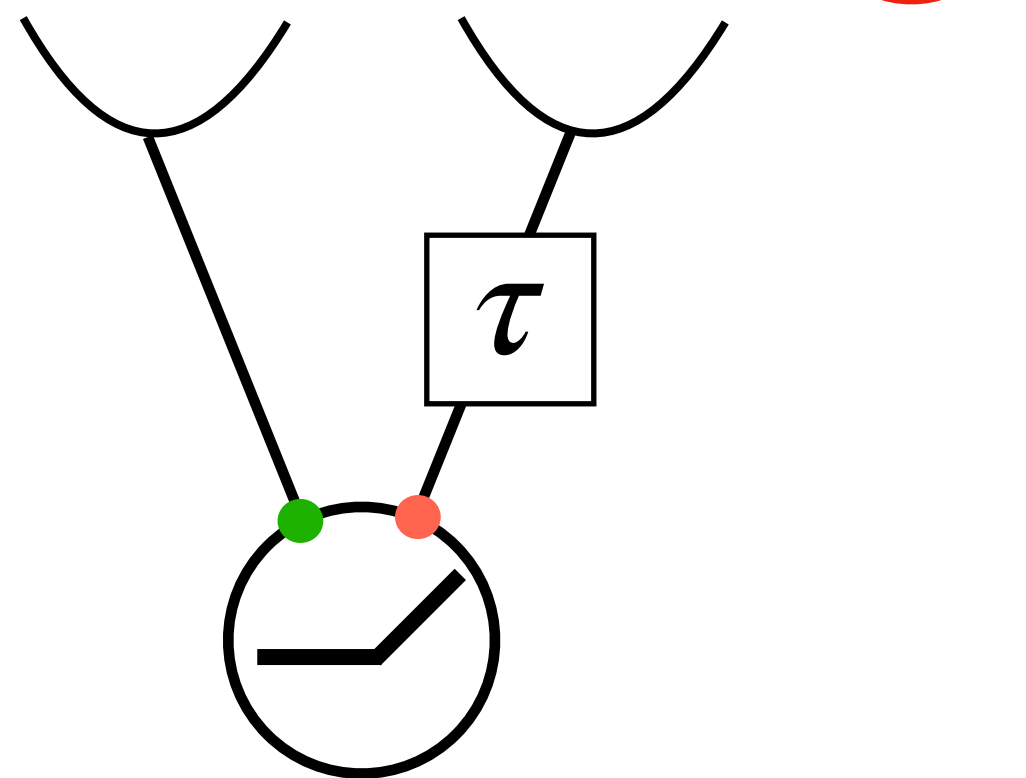


barlow-levick model

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direction

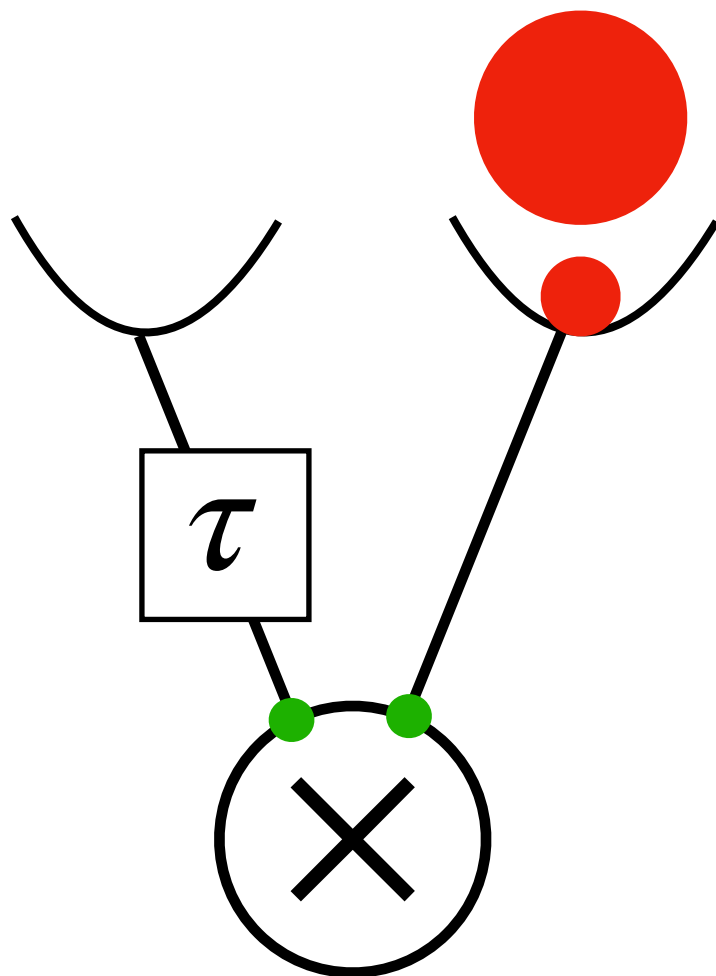


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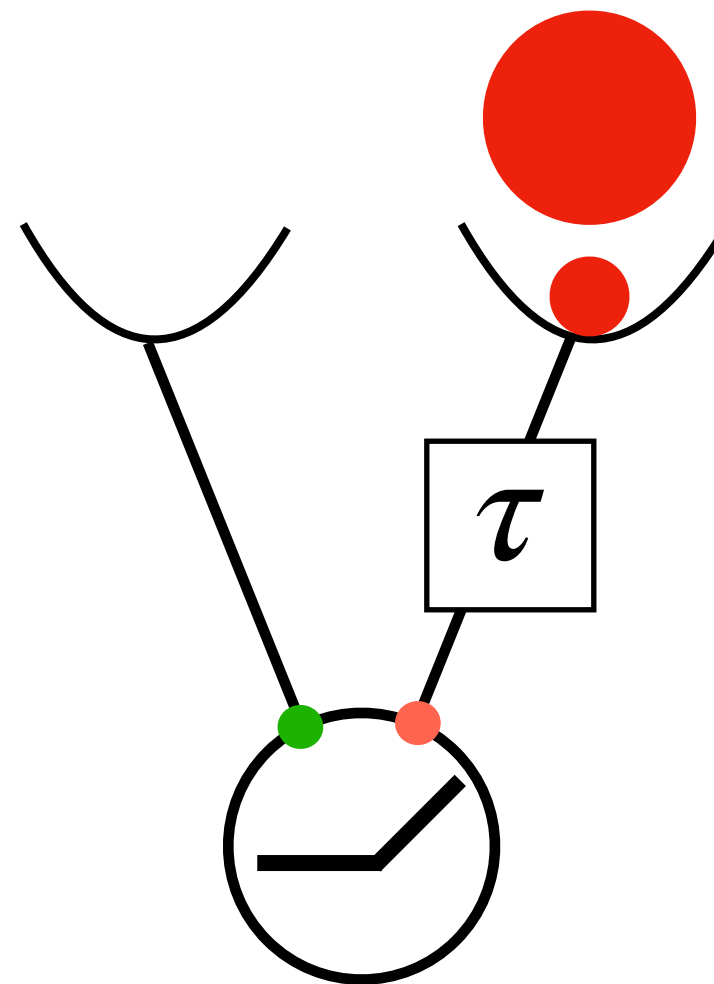


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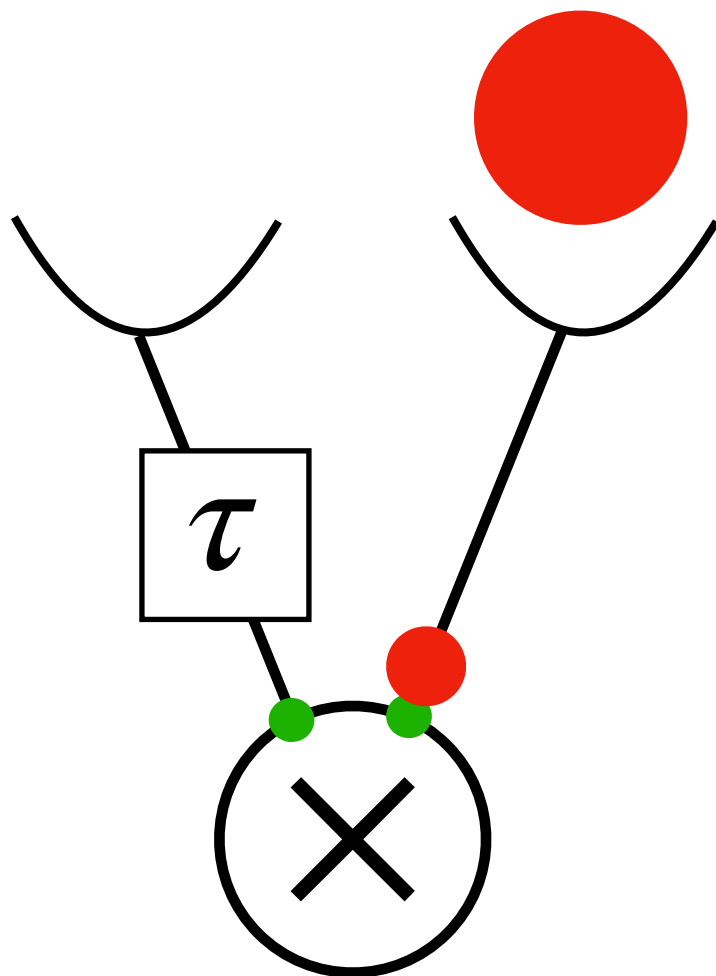


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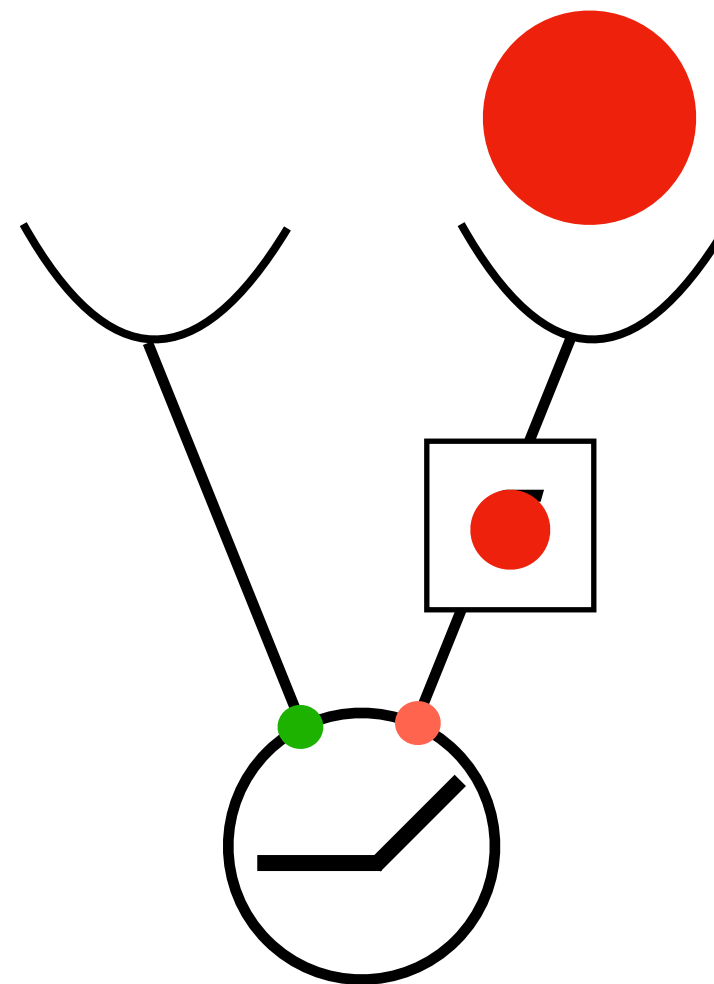


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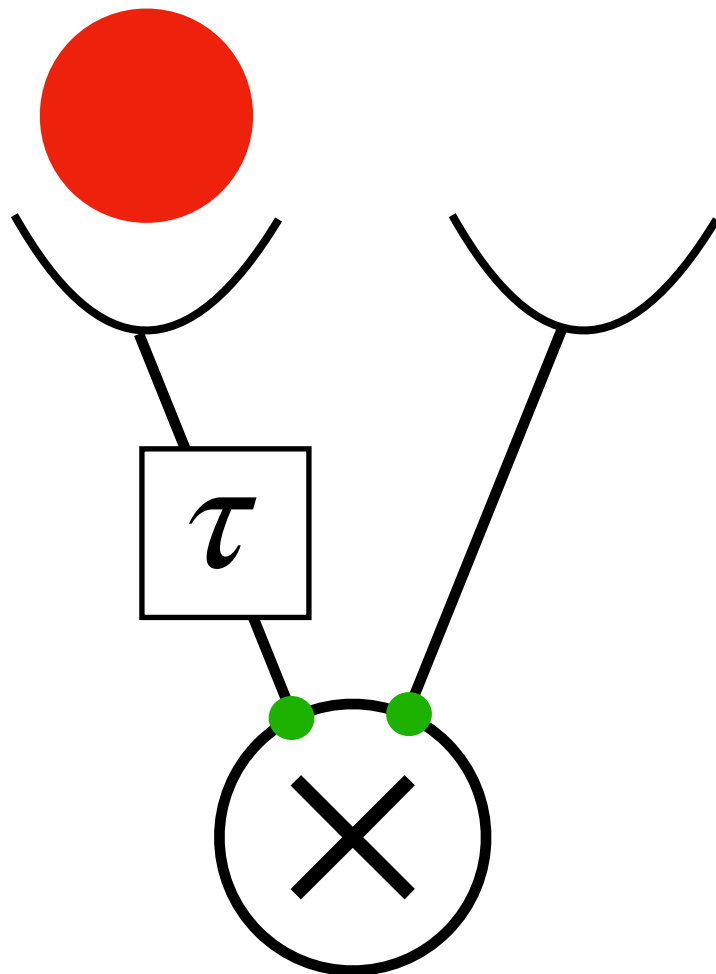


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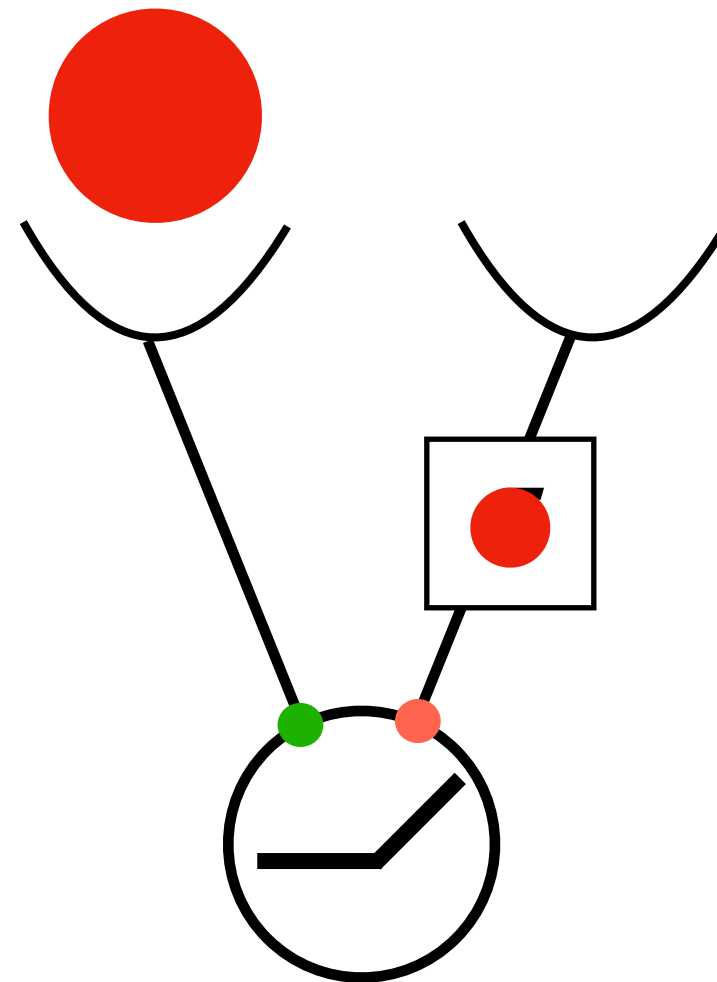


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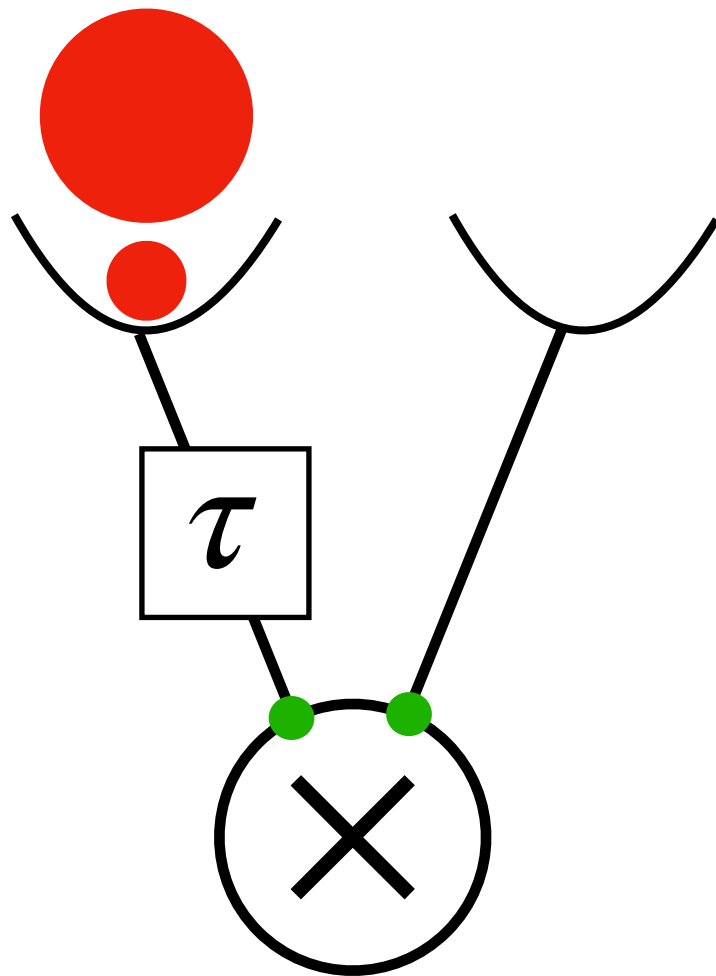


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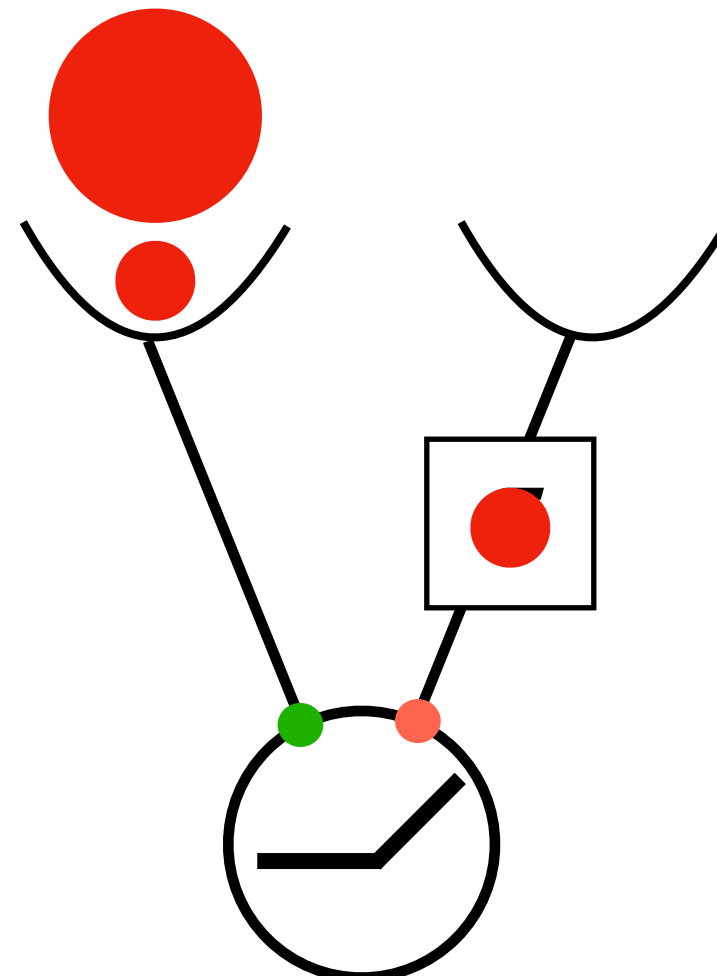


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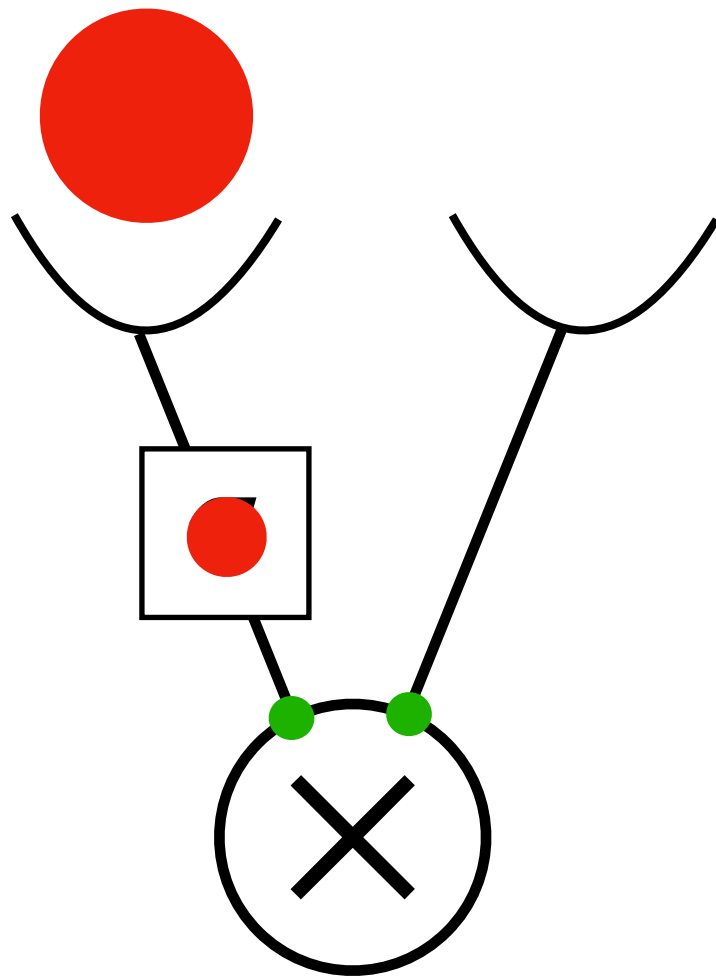


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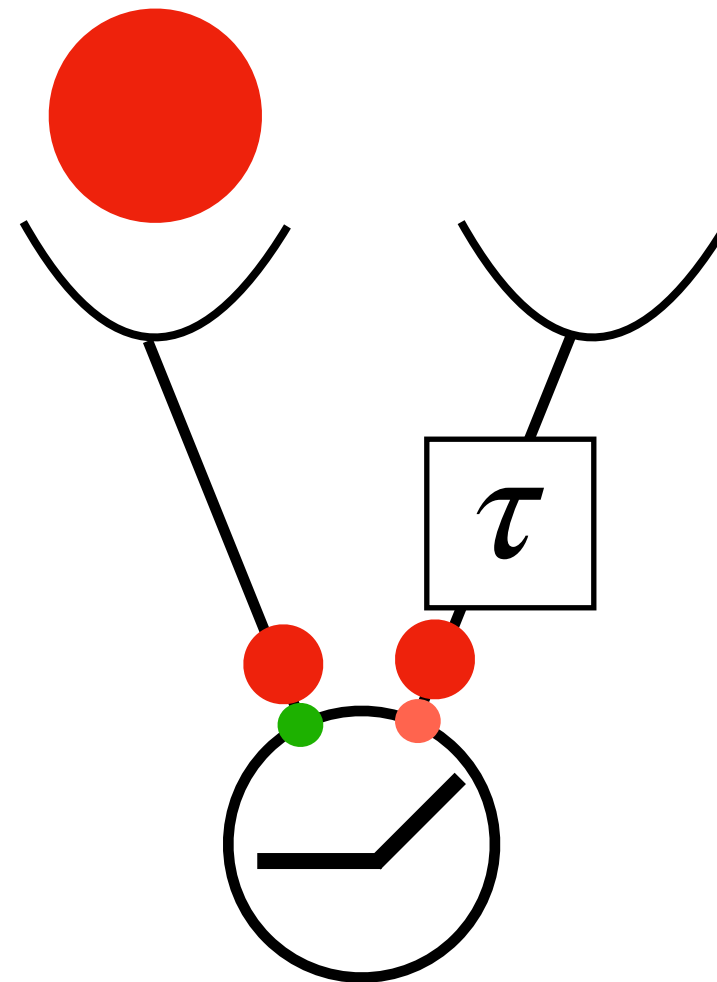


barlow-levick model

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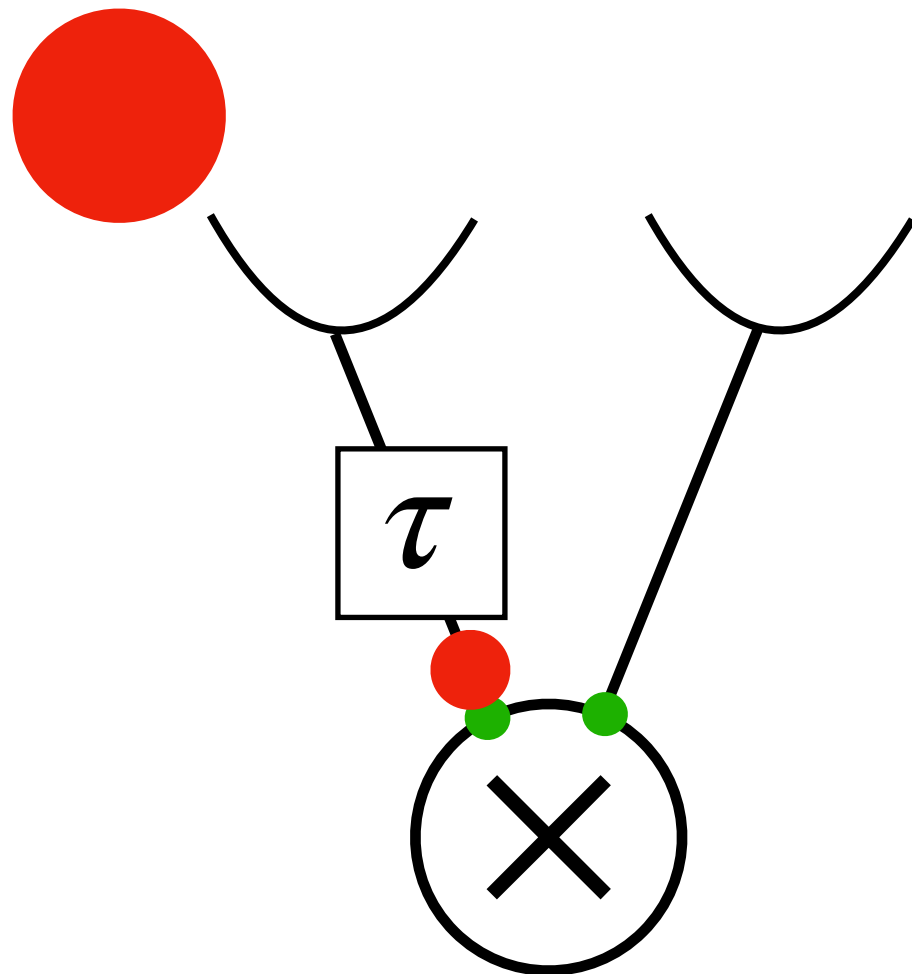


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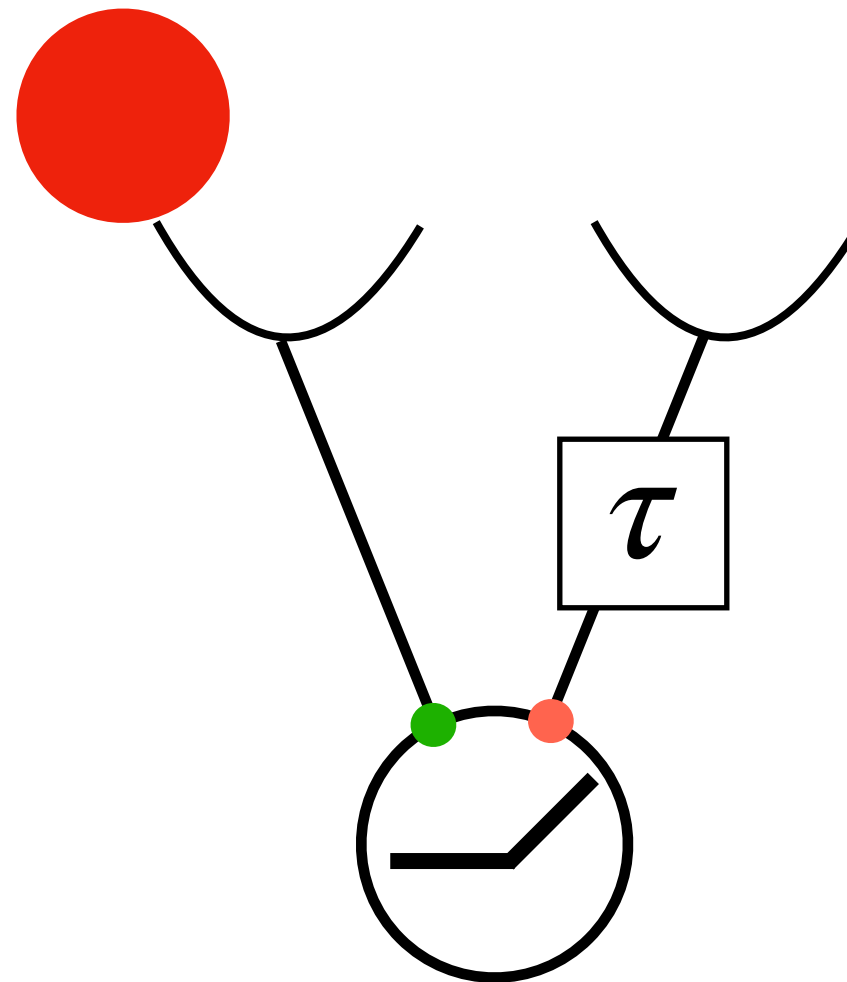


barlow-levick model

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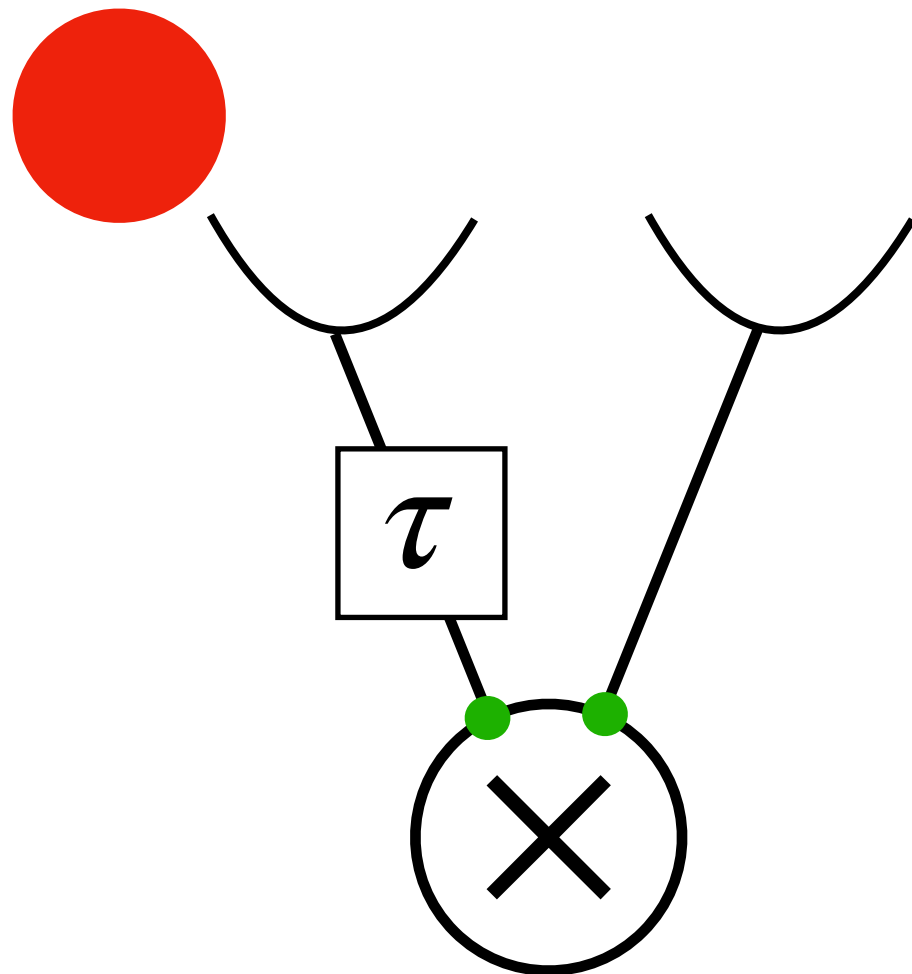


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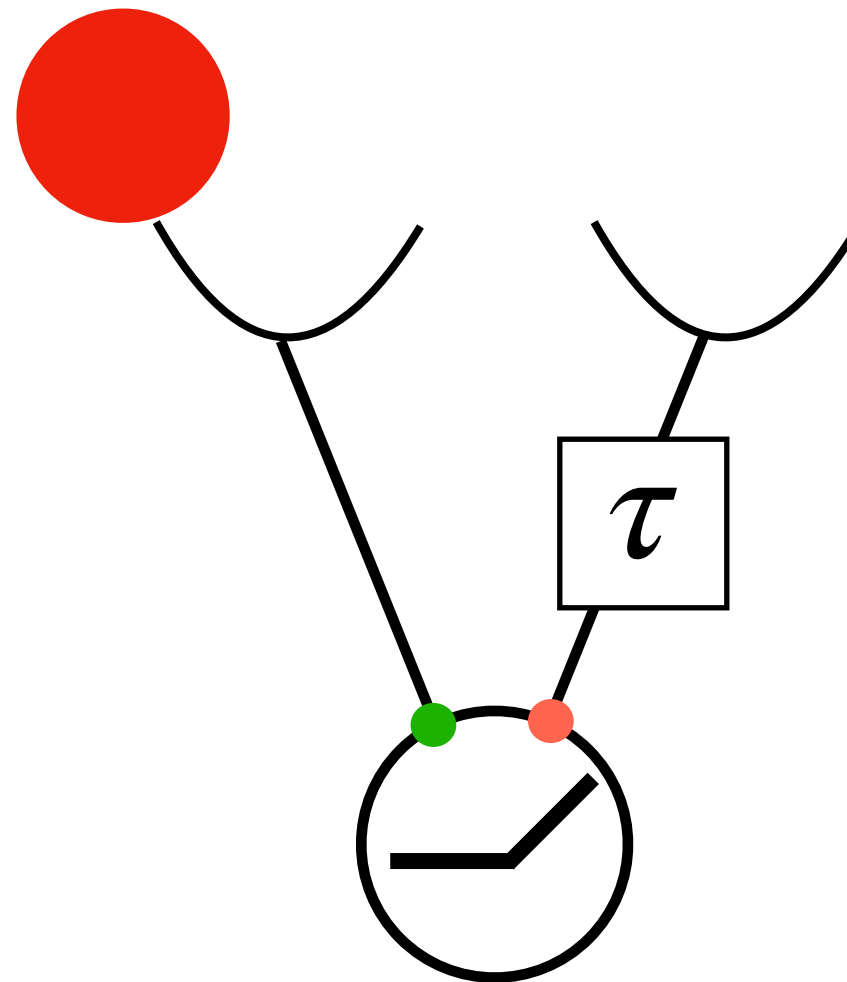


barlow-levick model

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direction



Reichardt model

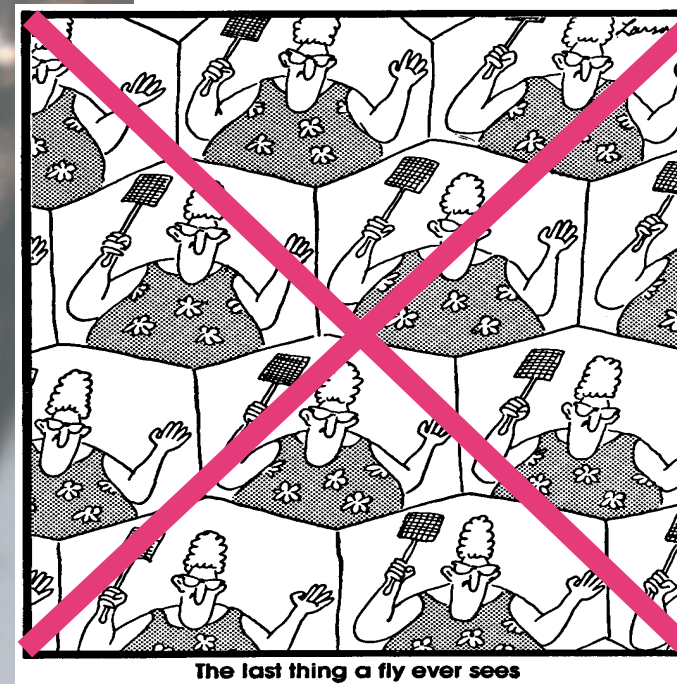


barlow-levick model

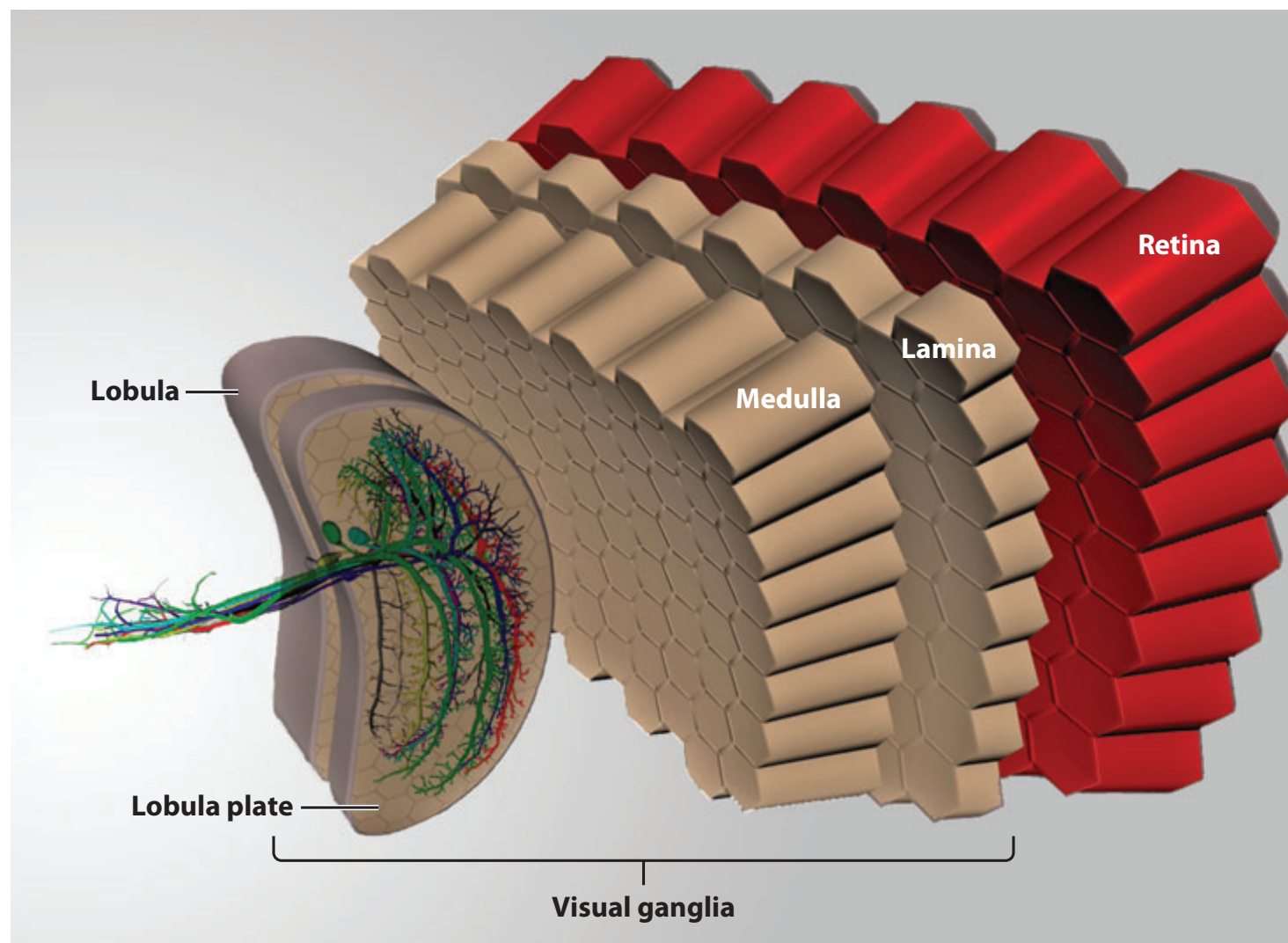
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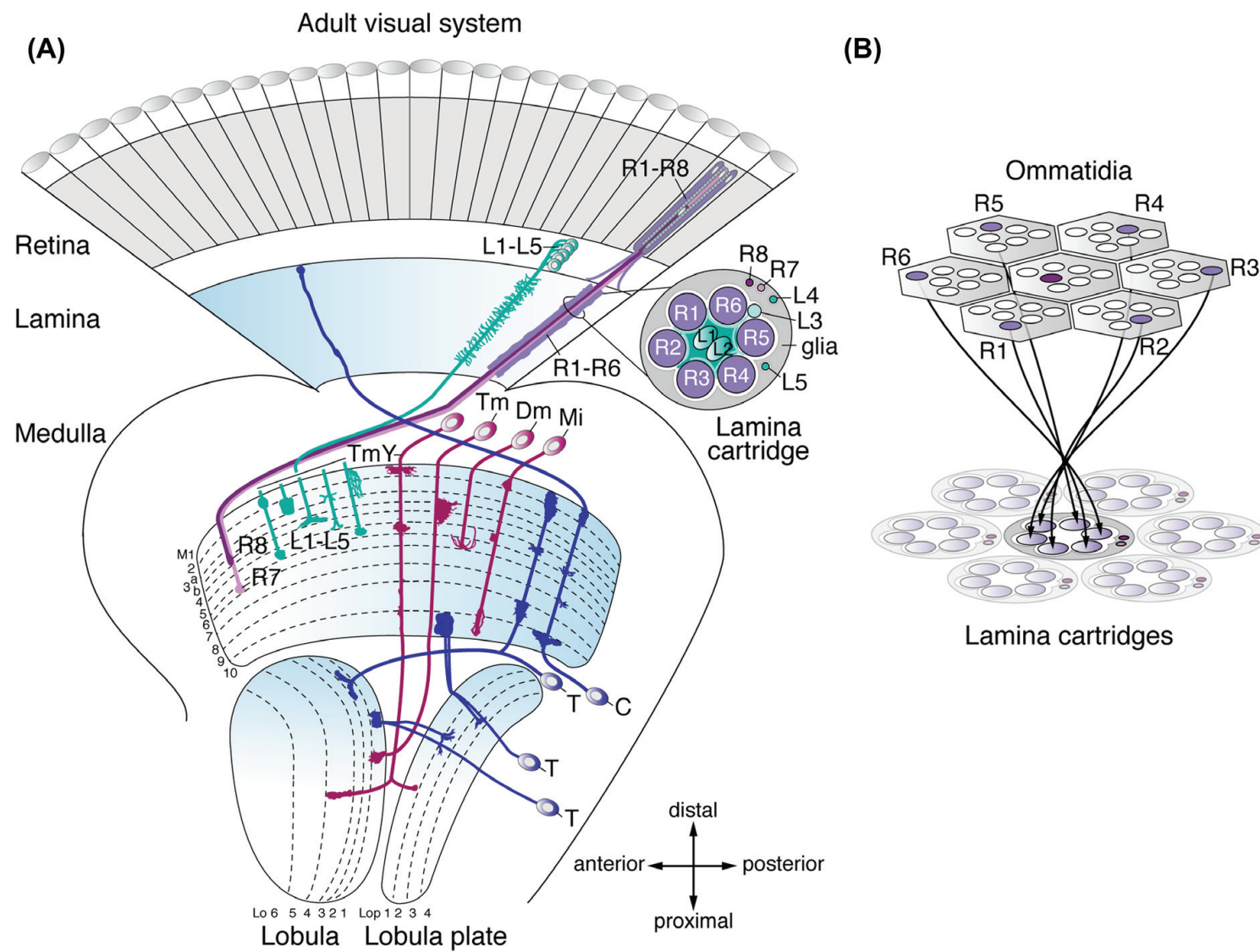
The Fly Visual System

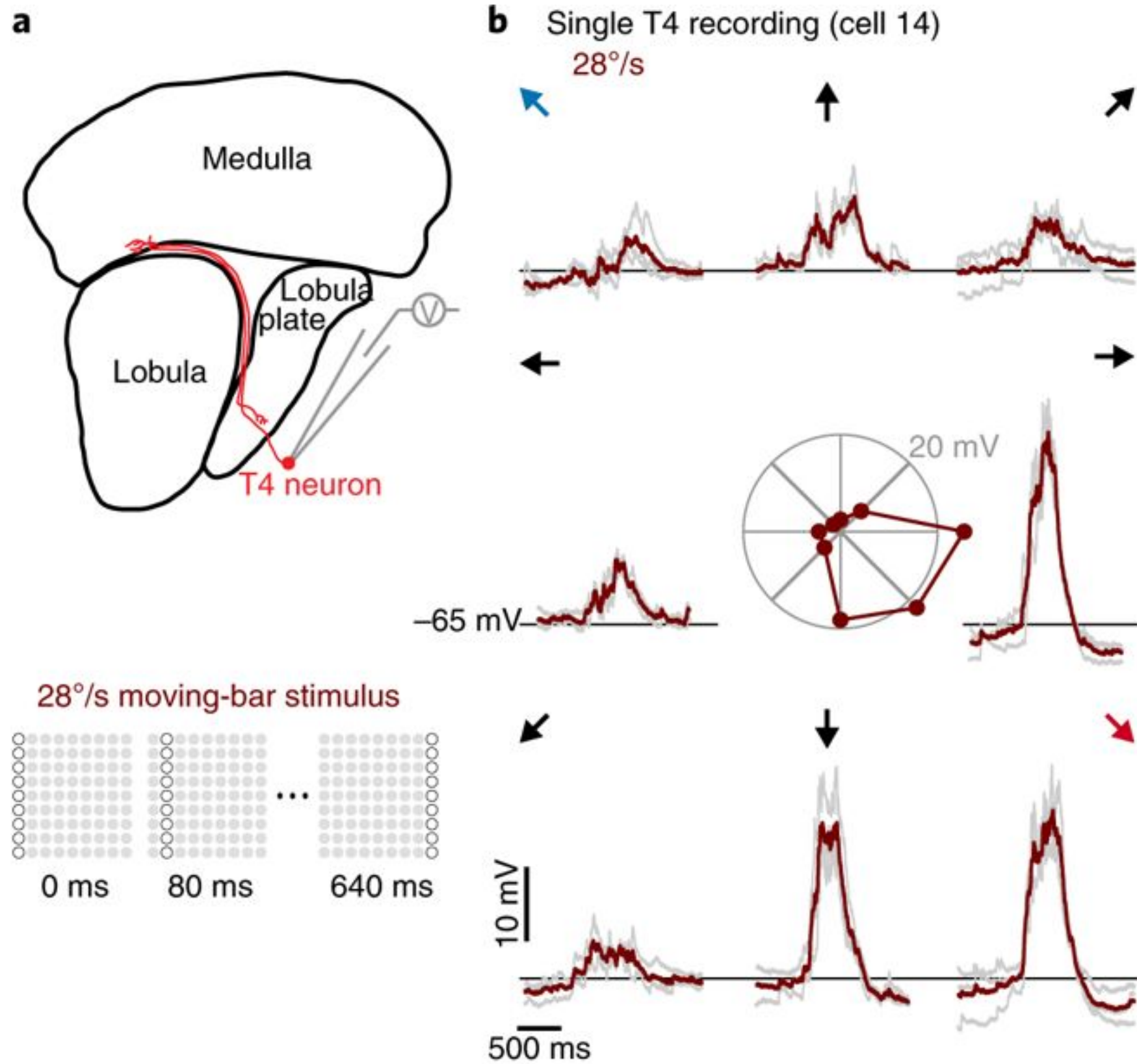


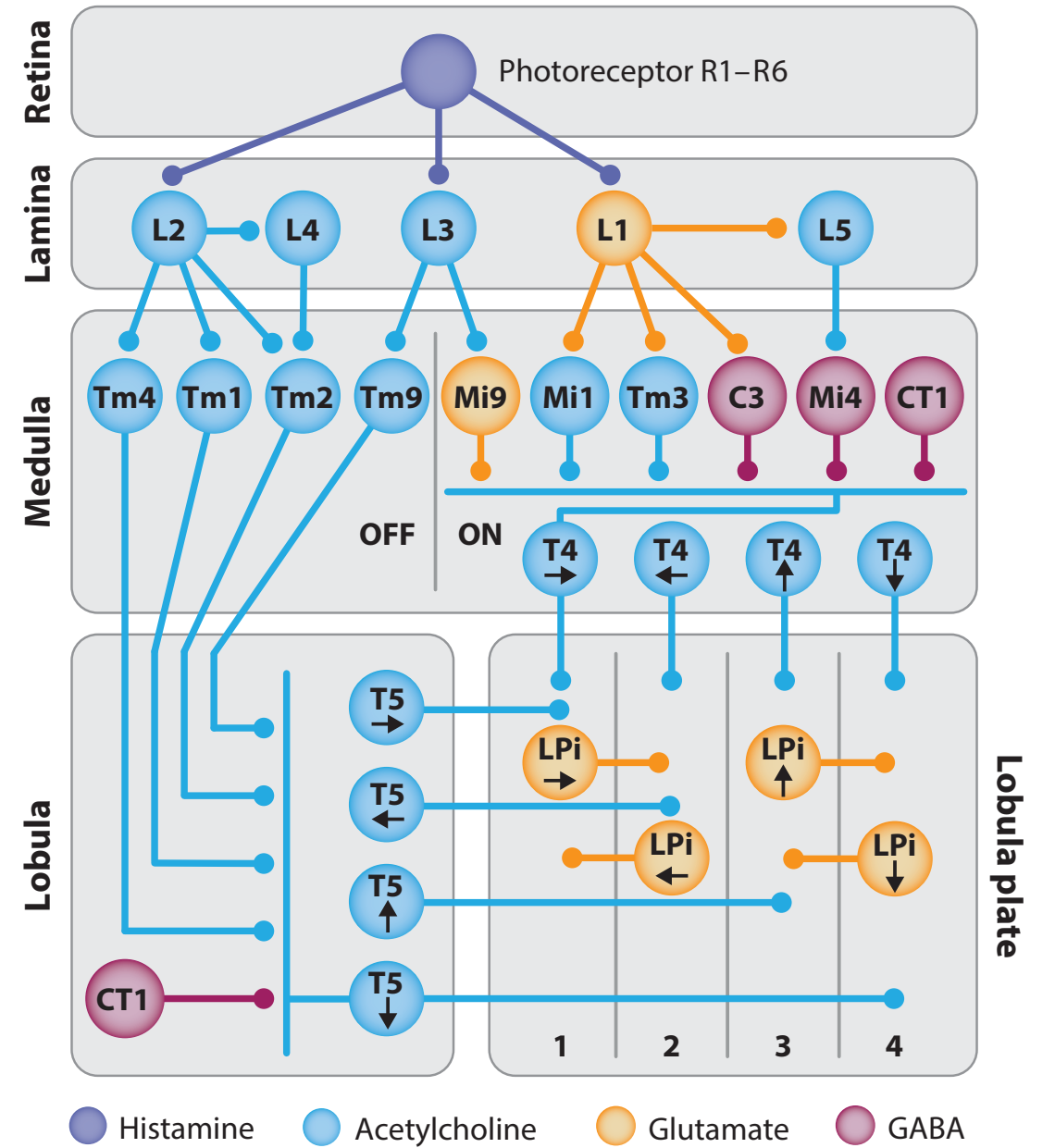
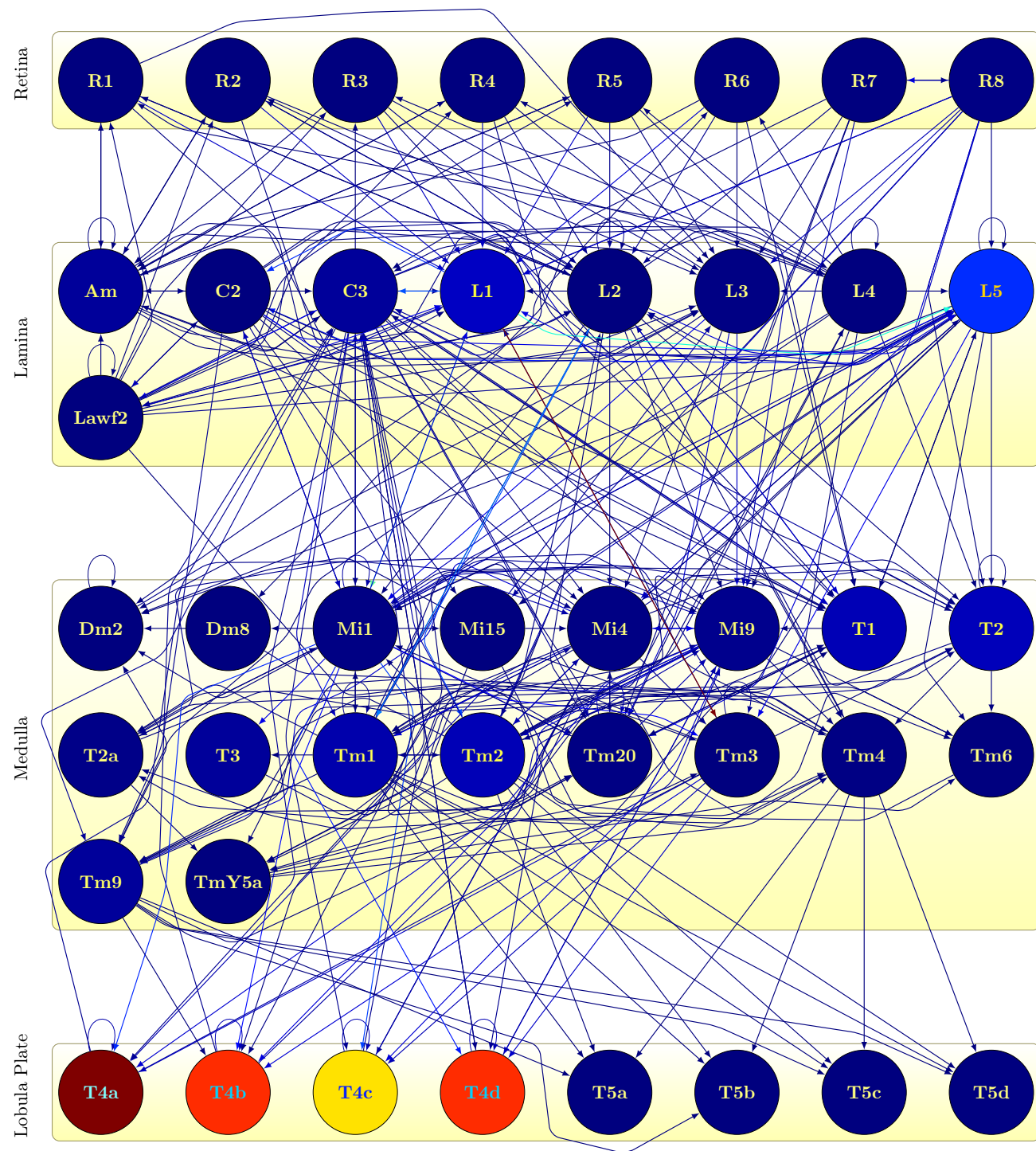
The Fly Visual System

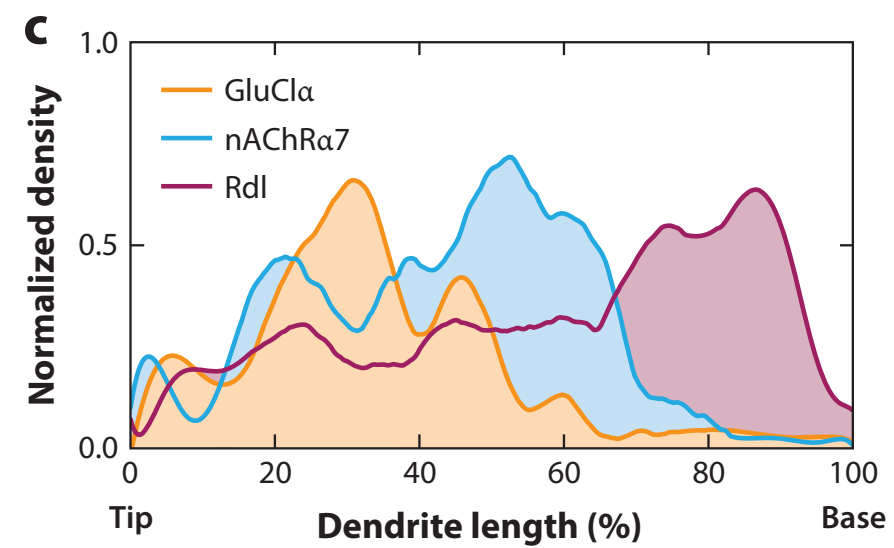
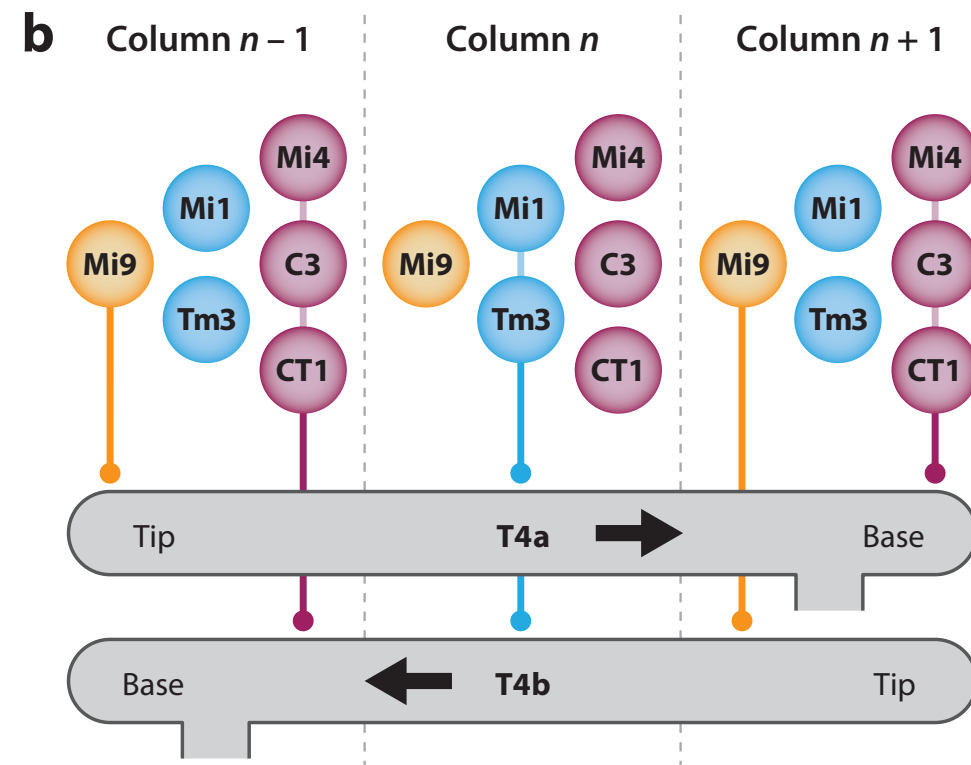
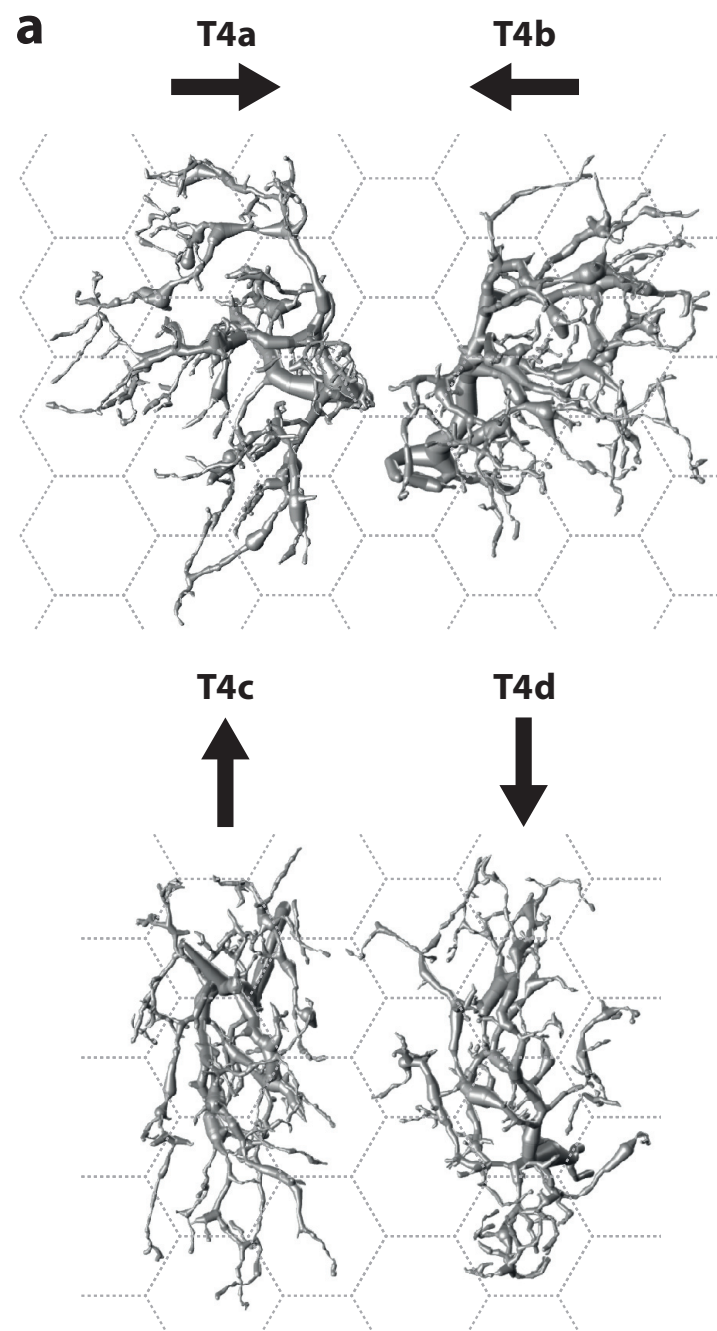


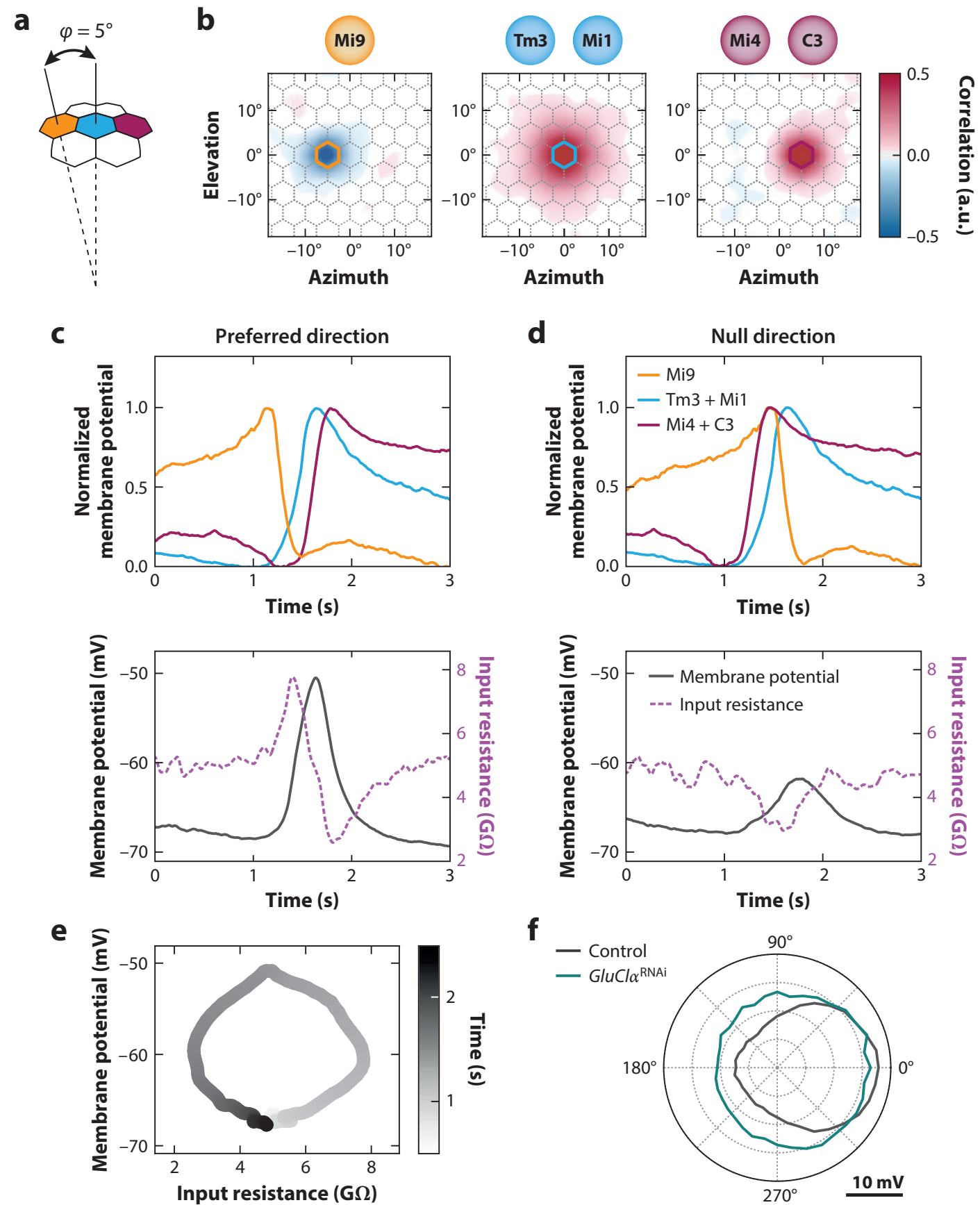
The Fly Visual System

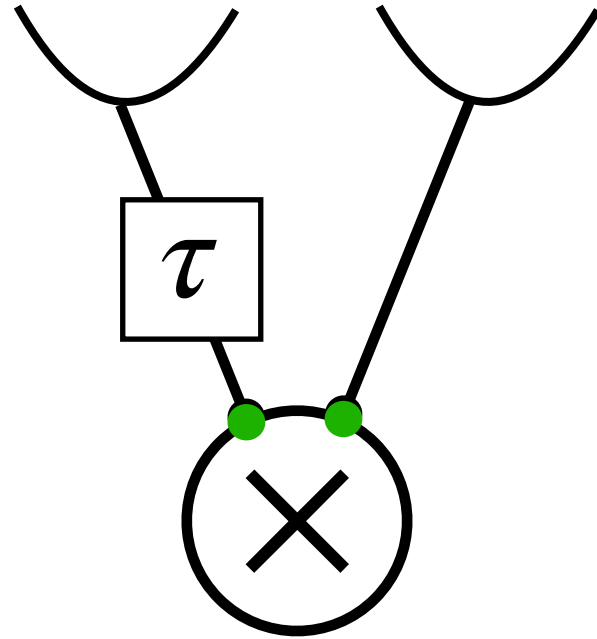




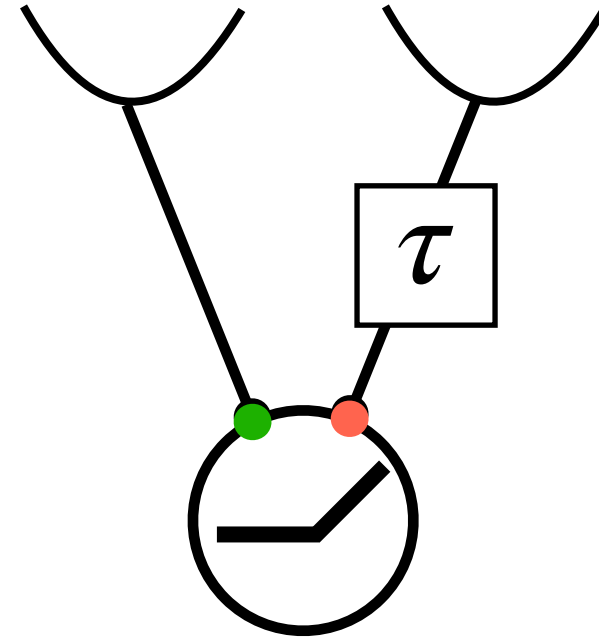




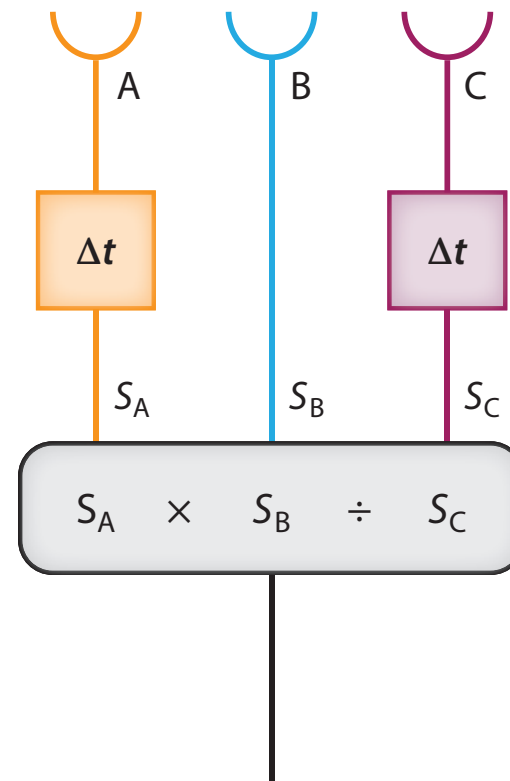


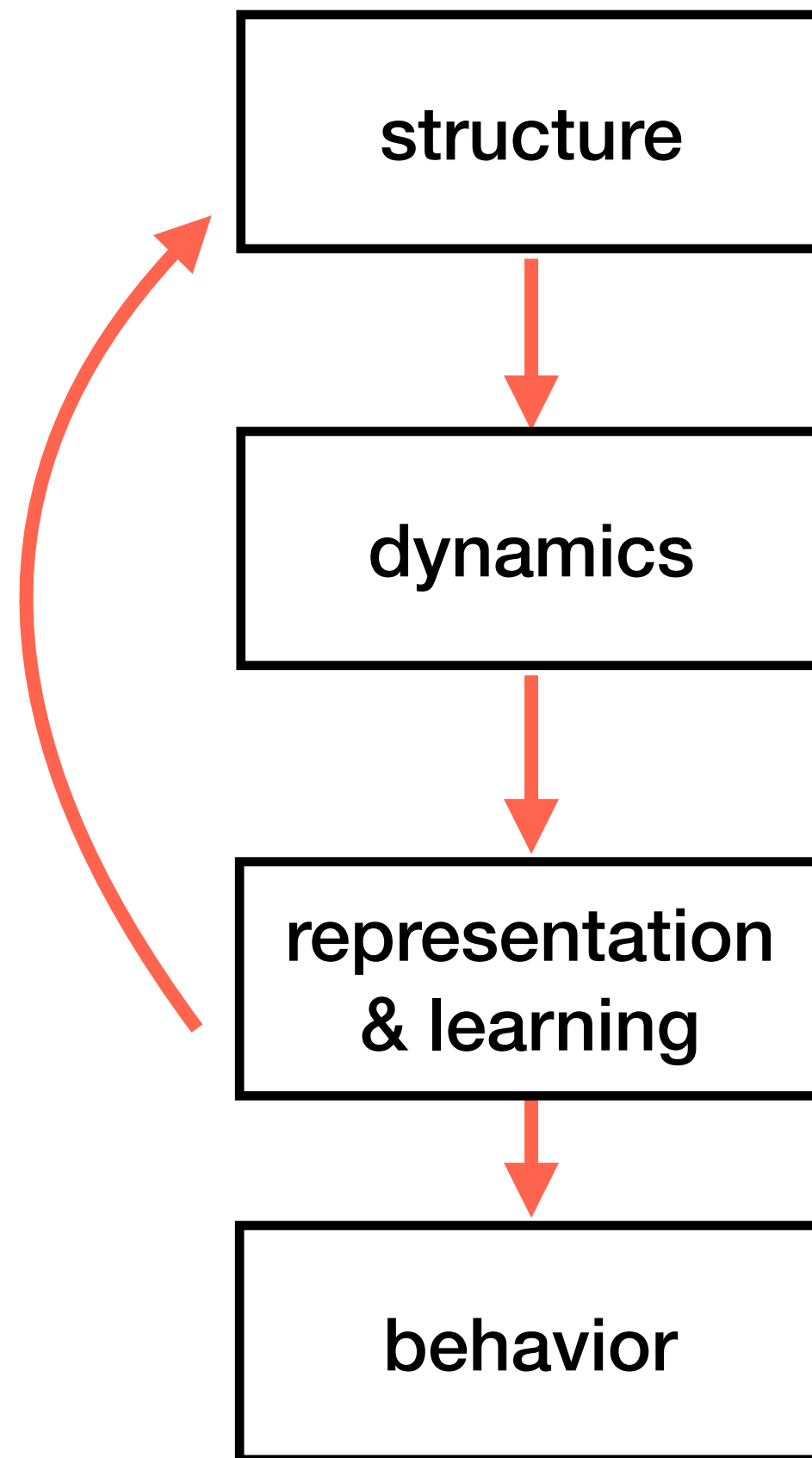


Reichardt



barlow-levick





Course Curriculum

- Introduction to Computational Neuroscience
- Single Neuron Computation
- Connect Neurons to Perform Computation
- Recurrent network and Attractor Dynamics
- E/I balance and Chaotic Dynamics
- Efficient coding principle
- Stimulus discrimination, Fisher information and Information-limited correlation
- Normative approach to neural computation and the emergence of sensory representation
- Dimensionality reduction and motor representation
- How does an animal move, an integrated approach, and insights from *C. elegans*
- Reward and Reinforcement Learning